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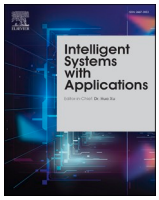

Leak detection and localization in underground water supply system using thermal imaging and geophone signals through machine learning

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ABSTRACT

The underground water pipeline system is a crucial infrastructure that largely remains out of sight. However, it is the source of a clean and uninterrupted flow of water for our everyday lives. Various factors, including corrosion, material degradation, ground movement, and improper maintenance, cause pipe leaks, a silent crisis that causes an estimated 39 billion dollars of loss every year. Prompt leakage detection and localization can help reduce the loss. This research investigates the potential of two machine learning models as supporting tools for surveying extensive areas to identify and pinpoint the location of underground leaks. The presented combined approach ensures the speed and accuracy of the leakage survey. The first machine learning model is a hybrid ML model that employs thermal imaging to identify subterranean water leakage. It relies on detecting thermal anomalies and distinctive signatures associated with water leakage to identify and locate underground water leakage. The developed model can detect up to 750 mm underground leakage with 95.20 % accuracy. The second model uses binaural audio from geophones to localize the leakage position. The model utilizes interaural time difference and interaural phase difference for localization purposes, and the 1D-CNN network delivers an angle in twenty-degree increments with an accuracy of 88.19 %. Large-scale implementation of the proposed model could be a powerful catalyst to reduce water loss in the water supply system.

1. Introduction

Despite rapid progress in many fields of science, the water crisis is still a global issue in the twenty-first century. Safe drinking water is a fundamental human right. However, according to the United Nations, over 2 billion people lack access to safe drinking water, leading to an increase in waterborne diseases and deaths worldwide (Goal, 2022). On the other hand, according to UNICEF, more than 1.42 billion people live in areas of high, or extremely high, water vulnerability in the world. Out of 1.42 billion people, 450 million are children, highlighting that one in five children lives in water poverty (UNICEF, 2021).

Water leakage is one of the leading causes of water shortage. According to Liemberger et al., non-revenue water, which is the water that is produced but not sold or used, has a value of 39 billion US dollars annually (Liemberger & Wyatt, 2019). According to Zhang and Wang, the European Union loses an average of 24 % of its supply of water due to leakage, in the United States the number is 12 %, and in Australia it is 10 % (Zhang & Wang, 2017). The economy and public health experience a ripple effect due to the water loss. Prolonged water leakage can lead to structural harm, sinkhole formation, and contamination of the water supply system with disease-causing pathogens. Moreover, the leakage contributes to a decline in the quality of water. Water loss results in reduced revenue, leading to increased water expenses and decreased service quality.

Researchers used different techniques to detect and locate water leakage quickly and efficiently. Trutié-Carrero et al. used a piezoelectric sensor and dyadic transform algorithm to detect leaks less than 1 mm in diameter. The major problem with the research was that the sensor needed to touch the pipe to acquire the vibration data, and the sampling rate was relatively low at 500 Hz (Trutié-Carrero et al., 2023). We showed that the leakage signal could produce frequencies up to 4.5 kHz, so a lot of information was absent at low sampling frequencies (Islam et al., 2023). Durai et al. developed an inspection robot with a magnetic flux sensor to monitor pipeline health. It used a permanent magnet and giant magnetoresistance (GMR) array to detect flux leakage. A major drawback of the research included emptying the pipeline before

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entering the robot, and it was unusable in nonmagnetic materials like plastic pipes (Durai et al., 2022). Guo et al. used multiple fibre-optic hydrophones and a revised cross-correlation algorithm to detect and locate leakage in the pipe (Guo et al., 2022). The downside of using fibre optics was that it was quite expensive, and the device only works with metallic pipes.

In 1940, the infrared camera was first invented. Researchers are using this technology for various applications, from medical diagnostics to detecting thermal leakage in houses. Infrared (IR) thermography is a promising field, but the application of IR thermography in water leakage detection has yet to gain high traction. The earliest use of the thermal camera in leak detection could be traced back to 1980. Eidenshink (1985), Weil and Graf (1991) and Weil et al. (1994) discussed leak detection trials in the 1980s. Their study could detect buried pipes, erosion and the void surrounding it. On the warmer surface, the leak appeared as a cool spot. The temperature difference between the leaked region and regular ground was five degrees Celsius. The main benefit of this technology was that it was a passive form of leakage detection technology.

Hu et al. summarized the leakage location technologies, and they showed that the most common methods for leakage localizations were cross-correlation, wavelet packet decomposition, empirical mode decomposition, beam forming, and linear prediction (Hu et al., 2021). Santos-Ruiz et al. used pressure data and a machine learning model to locate leakage (Santos-Ruizm et al.,). Tornyeviadzi et al. used SCADA data and a 1D-ANN network to localize leaks within a three-hundred-meter radius (Tornyeviadzi & Seidu, 2023).There are some other ways, including pressure and flow sensors (Alves Coelho et al., 2020; Gericke & Kuriakose, 2020; Lin et al., 2020), accelerometers (Li et al., 2020; Liu et al., 2019; Xu et al., 2022), GPR (Catapano et al., 2022; Yang et al., 2022), hydrophones (Phua et al., 2020), and tracer gas (Ghazi & Marshall, 2014) are some of the methods. One of the major disadvantages of the research mentioned above is that all of them require physical contact (excluding GPR) of the sensor for data collection. It limits their usability in underground and hard-to-reach pipelines. In this context, infrared thermography and geophone could pave a new way for leakage detection research.

The main benefits of thermography and geophone are that these are non-contact and non-invasive methods. It is less disruptive and does not require earth or building material removal. It is quicker than other techniques as it only requires capturing some images and audio data without disturbing the areas and is suitable for surveying large areas. It does not require any chemicals when capturing data, so it is much safer than tracer gas methods (Ghazi & Marshall, 2014). Though thermography and geophone are beneficial in water leakage detection and localization, this field lacks robust and accurate ML-based algorithms.

1.1. Motivation of the study

Water supply pipes are embedded in the cities and suburbs. These underground embedded pipes cover large lengths. According to the American Society of Civil Engineers it is estimated that there are 2.2 million miles of underground pipes that deliver safe, reliable water to the consumers (ASCE). In Europe, the estimated length of the drinking water network is 4.3 million km (Eureau, 2021). It is important to survey the water network regularly to detect, locate and stop the leakage. If a small leak is undetected for a long time, then that will cause a large amount of waste. Quickly and accurately surveying all the pipelines regularly in an area is essential for conserving the resource.

We adopted two-step approaches for water leakage detection and localization for speed and accuracy. In the two-step approach, thermal imaging is used for speed, and geophone is used for ensuring accuracy.

The first step is thermal image-based surveying. This would give an approximate location of the underground water leak. As soil is a non-homogenous material, the leakage location detected from the thermal images could be off as water could take curved paths to reach the surface. The temperature signature of the surface where the leakage is observed may differ from the actual leak location. So, in the second stage, geophones needed to be used to triangulate the precious leakage location. Geophones should be placed in at least two locations near the interest area. It would record the leakage noise. Based on the leakage noise, interaural time difference (ITD) and interaural phase difference (IPD) could be calculated and fed into the ML model to get the angle of the leak location relative to the geophone location. The intersecting point of the angles could be treated as the possible leakage location.

In real-life scenarios, the system could detect and locate leakage in a large area like a suburb. A drone or a car could be programmed to take IR images along the supply line or alongside the road to capture the IR images. Then, the thermal ML model could spot the potential leakage location. Thermal images of the same location needed to be captured multiple times at various times of the day. The multiple images would help to remove unwanted wet patches like wet leaves due to fog. In the second step, the geo-phone-based model could be used to triangulate the location to find the exact location of the leakage.

A thermal image-based model could be used to survey a large area in a short time, but accuracy could suffer because of its reliance only on the thermal signature. For example, the system could falsely detect a puddle as a leakage because of the puddle's thermal signature. The second step verified the leakage based on water leakage noise. It also provided an underground leakage location. The two steps are presented as two different methods in this study. The two steps complement each other.

1.2. Research objectives

This research aims to advance leak detection and localization in underground water supply systems by integrating thermal imaging and geophone signals. The goal is to develop two ML models. The first one would detect leakage based on thermal images and provide an approximate location of the leakage. The second model would use geophone audio signals to pinpoint the possible leakage location.

1.3. Research contribution

In this research, we set up two pilot projects to make a thermographic dataset and another audio dataset of leakage. The first pilot project captured the thermal variation of soil for underground leakage. The soil depths over the leakage were varied to capture different real-life conditions. The captured images were categorized in a dataset. Then, the dataset was used to train an ML model to spot the temporal variation of the soil to detect leakage. In the second pilot project, geophones were used to capture leakage audio from a variety of angles. Sensor location, soil depth, and audio pressure were changed to capture a wide range of underground leakage signals. A machine learning model was trained using geophone audio data to indicate the location of leaks. The model displayed the pan angle in relation to the geophone.

1.4. Research organization

The rest of the research is organized as follows. Section two, titled "Background," discusses different aspects of thermography and geophones in leakage detection and localization research. Section three is the "Literature Review" section, which discusses previous research and its limitations. The "Methodology" section is described in section four, outlining the whole research. This section gives an overview of the complete research and describes their relationship. Two experiments are discussed here. The first is "Underground Water Leak Detection Using Thermal Images", and the second is "Underground Leak Localization Using Geophones." Section five, "Underground Water Leak Detection Using Thermal Images", describes the full experiment setup conditions for data capture, augmentation, and image processing. The ML model section describes all the tested models and the best ML model architecture. The thermal image-based model includes data augmentation,

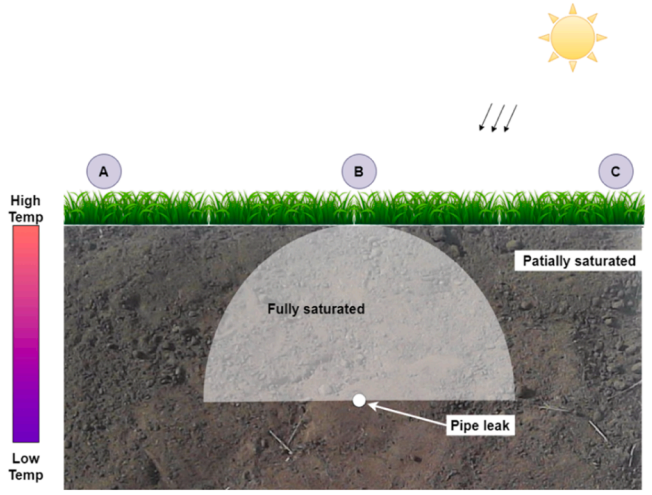


Fig. 1. Conceptualizing the hydraulic and thermal behavior of a buried pipe leak.

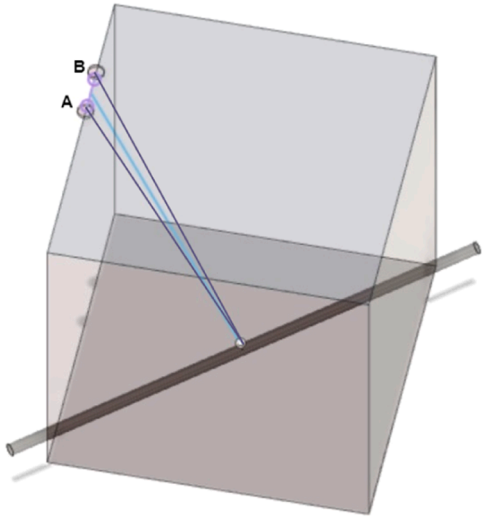


Fig. 2. Underground leakage sound source localization.

image preprocessing, feature extraction, model architecture, model performance, and model optimization. The following section, section six, describes the leakage localization process using geophones. The audio data capture, feature extraction, model development process, transfer learning, and model architecture are detailed here. The limitations of the research were discussed in section seven. Finally, the study concluded by discussing the results in section eight, which was named "Conclusion and Future Work."

2. Background

Soil is a complex system. Due to soil composition variance, it is not easy to establish a unique thermal conductivity or sound conductivity formula. Zhang et al. and Alrtimi et al. showed that the degree of saturation, dry density, soil mineralogy, particle size, gradation, and packing geometry would affect the thermal conductivity of the soil (Alrtimi et al., 2016; Zhang & Wang, 2017). They also observed that water and quartz content had a more prominent effect on thermal conductivity than other factors. Brutsaert et al. showed that the speed of sound in shallow depths of unsaturated soil was proportional to the 1/6 power of the effective pressure of soil (Brutsaert & Luthin, 1964). Albers studied the effect of soil saturation on longitudinal waves and shear waves and concluded that the velocity and attenuation of the wave depended on the initial soil saturation and wave frequency (Albers, 2009).

Peter and Jayantha, in their research on the “Reliability of infrared thermography in detecting leaks in buried water reticulation pipes”, proposed a simplified thermal interaction model (Bach & Kodikara, 2017). The soil’s thermal behavior was dependent on the hydraulic process inside the soil. The water flow in the soil was described using Darcy’s law. The hydraulic gradient would dictate the water flow when there was a leak. They assumed the water pipe would have higher pressure and the flow would occur from the leakage point to the outwards. They also assumed that the soil was not fully saturated. The water would have two types of flow. Downward flow was due to gravity and water pressure, and upward flow was due to capillary rise and water pressure. The water flow rate in the soil is called hydraulic conductivity (k_{hyd}). The leaking water would disperse through the soil matrix. It would take the form of a bulb (wetting front) in all directions, as shown in Fig. 1.

The soil would have two zones. One saturated zone was inside the bulb and the other unsaturated or partially saturated zone was outside the bulb. If the water pressure stayed the same for a long time, then the wet zone of the bulb may reach the surface. Based on these assumptions, a clear understanding of the thermal behavior of soil could be visualized. A further investigation of thermal imaging for leakage detection could be undertaken and some assumptions like the rate of cooling and the way the vegetation affects the thermal imaging could be considered. When developing an intelligent model, all these considerations could be taken into account to make a robust model.

When leak sound propagates through soil media, the time, phase, and level difference between the two sensors would be different due to the distance difference between the sources. With binaural audio, the distinctive cues of two sensors could be used for sound source localization. Fig. 2 shows the setup for underground sound source localization. We could see the distance between the leak location of the pipe and the sensors on the top of the ground. The A and B denote the sensor locations, and there is a distance difference between the two sensors from the leak location. The binaural cue or feature could be extracted from the stereo audio to train ML models for leakage localization. The sensor pair could be setup on multiple points around the possible leakage location to triangulate the exact leakage point.

3. Literature review

Zhou et al. investigated the optimal circumstances for utilizing infrared thermography for earth dam leakage detection. IR images were further studied to explore the spatial distribution and temporal varying regularity of the images. According to the author, the transition time of the sun (i.e., during the morning and the evening) gave high contrast IR images, and the different thermal capacities of the soil and water heats and cools down at different rates (Zhou et al., 2022). In another work, Zhou et al. studied the feasibility of locating the submerged piping effect in embankments using infrared (IR) images (Zhou et al., 2022). Yahia et al. used image processing using multitemporal IR analysis. Luminance transforms were used on the images to remove the effect of time variation, and temporal variation image (TVI) was computed. After that, a threshold was set on the TVI histogram to separate the leak and non-leak areas (Yahia et al., 2021). The researchers could distinguish between shadow, non-leak, and leak areas from one IR image. One drawback of this research was that the authors needed to provide more information regarding the accuracy of their work. Aslam et al. conducted studies about the usability of Ground Penetration Radar (GPR), spectrometers, and infrared (IR) cameras. IR camera and GPR were most reliable at detecting leakage at 2 % and 5 % moisture content. IR cameras had 66.57 % to 100 % accuracy on leakage detection (Aslam et al., 2022). One drawback of the research was that when the moisture content was equal to or more than 10 %, the algorithm did not work. In real life, the

moisture content of the soil would be unpredictable and uncontrollable due to leakage.

For buried pipes, Peter and Jayantha conducted a study to understand the reliability of detecting leaks in buried pipes using infrared thermography (Bach & Kodikara, 2017). They focused on small-diameter buried pipes (i.e., around 100 – 400 mm) with depths ranging from 0.8 to 1.2 m. The authors collaborated with the Melbourne water utility company and photographed known leakage locations with an IR camera. For this study, the authors photographed the exact location at different times of the day. It should be noted that the authors did not use any algorithm or machine learning techniques. They photographed the location with a thermal camera and a standard digital camera. They tried visually establishing a relationship between the leakage and the thermal image. They investigated twenty-seven sites and could identify leaks with 59 % accuracy, and 22 % of the sites had a notable but inconclusive thermal signature. They concluded that indicators like the surfacing of water, excessive vegetation, and surface migration of heat made detecting leakage easier. Unlike the previous research, they included all conditions, including vegetation and seasonal differences (Bach & Kodikara, 2017). Alaa Al Hawari, in his paper titled "Non-Destructive Visual-Statistical Approach to Detect Leaks in Water Mains", used IR images and statical analysis to identify leaks in the pipe on the ground surface. This method used the normal distribution curve to predict the leakage location. The researchers assumed that temperature distribution in dry conditions would follow a normal distribution curve. The thermal distribution of the two-tailing end ($T \geq Z$ (95 %)) would provide an indication of leakage in the pipe system. The identified leak location was visually inspected to verify the result. The authors claimed that the result yielded 95 % accuracy (Al Hawari et al., 2015). The drawback of the research was that it only considered dry weather conditions and did not consider humid weather conditions.

Carreño tried to detect buried pipes using thermal images and data mining. The researchers used KNIME (data analytic software) to develop a relationship between the buried pipes and thermal signatures. The main drawback of this research was that it did not provide a complete model of the process and the accuracy of their work. The authors suggested proposing a new and improved mathematical model in the future (Carreño-Alvarado et al., 2014). Shakmak et al. experimented with the effect of different-resolution thermal cameras in leakage detection (Shakmak & Al-Habaibeh, 2015). They used a controlled lab experiment to evaluate the possibility of using a low-cost thermal sensor to detect water leakage in the supply system. The researchers proposed using a helium balloon or drone to survey leakage in a larger area. The pipes were buried 11 cm deep. The researchers did not use an algorithm to detect the leakage in the IR images. They only used visual cues to determine the leakage. In 2018, Penteado et al. experimented with a digital image processing method with thermal images. The authors mainly used the q-sigmoid function to process the images. They could separate the leakage location from the other areas in their lab-based experiments. The researchers included both sandy and clay soil for the experiment, and at 10 cm pipe depth, this result can be used for garden or backyard leak detection (Penteado et al., 2018). Fahmy and Moselhi conducted extensive research in Montreal (Canada) about the factors affecting the applicability and limitations of IR cameras in leak detection. The researchers developed a model that detected leaks in the fall and spring but failed in the summer and winter. The model needed to be more generalized. The model was explicitly designed for pavements. Their model had parameters such as pavement surface radiation, moisture in the soil, solar radiation, heat transfer between the environment, wind speed and temperature. They assumed that the thermal contrast due to the water leak would take the shape of a cone, and the head of the cone would represent the approximate location of the leak. They compared their findings with the acoustic method; the minimum error was 1.01 m to a maximum of 2.30 m (Fahmy & Moselhi, 2010).

Bao et al. studied the application of machine vision in power plant water leakage detection. They used a wireless robotic thermal camera for image acquisition. A combination of Gaussian filter, top hat transform, canny edge detection, contour closure and image merging were used to develop their algorithm. The authors claimed the algorithm can detect surface water leakage with 99 % accuracy (Bao et al., 2022). Li et al. proposed a method based on active thermometry and fiber Bragg grating (FBG) based quasi-distributed fiber optic temperature. The researchers fabricated a special cable type with a heating element, quasi-distributed fiber optic temperature sensor, reinforcement element, and protective layer. This temperature-sensing cable needs to be placed alongside the water pipe. Temperature response was used to detect and locate water leakage in the system (Li et al., 2021). Atef et al. conducted more thorough research on leakage detection using IR images and ground-penetrating radar (GPR) (Atef et al., 2016). Their research included simulated leaks as well as verifying actual leak data. They

Table 1
Comparison of the algorithms and availability of the dataset.

Reference	Algorithm	Dataset
(Zhou et al., 2022)	Spatial distribution and temporal varying characteristics of thermal images of water leaks were studied. No algorithm was proposed.	Not available
(Zhou et al., 2022)	The algorithm scanned thermal images to pinpoint temperature disruptions caused by piping outlets. These appeared as cold or hot spots depending on time and weather. The Otsu method segmented the outlet, identifying the flow-affected area.	Not available
(Yahia et al., 2021)	The algorithm analyzed multiple IR images to track soil temperature changes over time. A luminance transformation was introduced to remove temperature variations caused by field temperature. Then, it created a temporal variation image (TVI). A threshold was set based on histogram analysis to segment leakage and non-leakage areas.	Not available
(Aslam et al., 2022)	Discussion of visual observation. No algorithm was proposed.	Not available
(Bach & Kodikara, 2017)	Discussion of visual observation. No algorithm was proposed.	Not available
(Al Hawari et al., 2015)	The researchers hypothesized that the temperature distribution under dry conditions would conform to a standard distribution curve. They inferred that the thermal distribution of the tail ends ($T \geq Z$ (95 %)) would signal potential leakage in the pipe system.	Not available
(Carreño-Alvarado et al., 2014)	A professional data analytics platform, KNIME, was used. The model was not described here.	Not available
(Shakmak & Al-Habaibeh, 2015)	Discussion of visual observation. No algorithm was proposed.	Not available
(Penteado et al., 2018)	The q-sigmoid function was used for preprocessing, and then a thresholding value of 0.5 was used to segment the image. Then, dilation was used to remove noise. Finally, classification was done based on region size and texture.	Not available
(Fahmy & Moselhi, 2010)	The authors described the fieldwork and the experimental protocol to investigate the factors that impact the use of thermal cameras for leakage detection use. They assumed that the leakage would form a cone shape from the source and developed a model to determine the approximate location of the leak.	Not available
(Bao et al., 2022)	At first, the region of interest was selected using contrast adjustment and image binarization. Then, a combination of Gaussian filter and top hat transform is used for effusion edge extraction. Then, canny edge detection is done to contour closure. The contour closures were used to detect effusion and leakage.	Not available

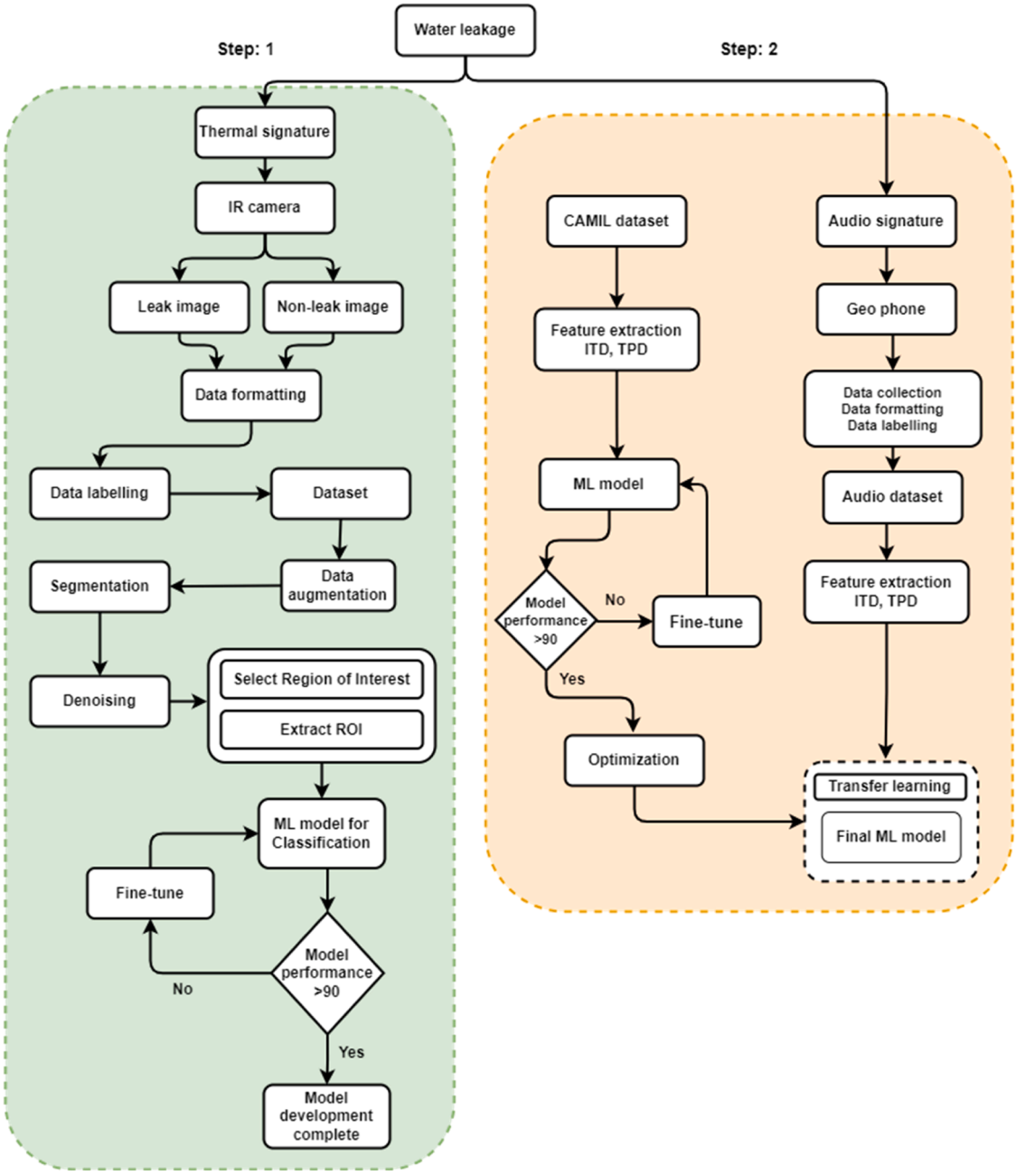


Fig. 3. Proposed methodology.

verified their result with ground-penetrating radar (GPR). Researchers used GPR to locate the buried pipes accurately and used the thermal image to pinpoint the possible location of the leak. They used a seeded region-growing algorithm on each IR image to differentiate the leakage and non-leakage areas. The centroid of the leakage area was then calculated using Green's theorem to pinpoint the leakage. They claimed their method can achieve accuracy up to 97 % by verifying their claim with GPR reading. It should be noted that the researchers used considerably low temperature feeding water for contrast in the thermal image. Table 1 shows the algorithm comparison and availability of the dataset. We can see that machine learning algorithms are absent in IR image-based water leakage detection works, and there is no publicly available IR dataset of water leakage detection for training ML algorithms as well.

Though there is no IR and ML-based research work for water leakage detection, Shi et al. used transfer learning and IR and ML-based approaches to detect hydrocarbon and gas leakage (Shi et al., 2020; Shi et al., 2023). Other temperature-based leakage detection systems include Wang et al.'s distributed temperature sensing method (Raman scattering) using sensing fiber (Wang et al., 2022). Awwad et al. used temperature variation between two points to detect underground water leakage (Awwad et al., 2020).

Geophones or ground microphones are listening devices that use sensitive piezoelectric materials to pick up leak-induced sound and vibration. Modern devices have amplifiers and noise filters to enhance the leakage signal. During the leak survey, the inspector regularly listens for leak noise from the surface to pinpoint the leakage location (Hunaidi & Wang, 2006). It is a slow and time-consuming process, and its effectiveness depends on the operator's experience (Hunaidi et al., 2000). Almeida et al. compared the performance of three types of sensors: hydrophones, geophones, and accelerometers (Almeida et al., 2014). The primary method used here was the cross-correlation of acoustic signals

measured by two sensors. The authors concluded that the pipe and the sensors acted as filters for the leak noise, affecting the bandwidth and the shape of the cross-correlation function used to estimate the leak position. They also proposed a quality measure for the data and concluded that the accelerometer performed the best in their test setup.

Grumiaux et al., in their survey about sound source localization (SSL), pointed out two main ways of SSL. The first was the conventional way, and the second was based on neural network architecture (Grumiaux et al., 2022). The conventional methods included direction of arrival (DoA) (Xue et al., 2020), beam forming method (Benesty et al., 2008), Subspace method (Benesty et al., 2008), Cross-correlation method (Knapp & Carter, 1976), Bayesian method (Gerstoft et al., 2018), and independent component analysis (ICA) method (Sawada et al., 2003). The deep learning-based methods include Feedforward neural networks (FFNNs) (Opochinsky et al., 2019; Pak & Shin, 2019), Convolutional neural networks (CNNs) (Bologni et al., 2021; Diaz-Guerra et al., 2020), Recurrent neural networks (RNNs) (Nguyen et al., 2021), Neural networks with residual connections (Naranjo-Alcazar et al., 2020), Neural networks with attention mechanisms (Adavanne et al., 2021), Encoder-decoder neural networks (Goodfellow et al., 2016). Grumiaux et al. also observed a general pipeline for deep learning-based SSL systems. Generally, multi-channel audio signals were used for feature extraction, and those input features were used to train the deep neural network for DoA estimation (Grumiaux et al., 2022).

Researchers used other innovative methods as well. Like dominant sensor sequencing methods (Li et al., 2022; Zhang et al., 2022), using optical fiber in conjunction with LED (Yang et al., 2022), using hydraulic modelling and leakage matrix (Fan & Yu, 2022), and using pressure and flow sensor data (Vrachimis et al., 2021).

From the research discussed above, ML algorithms have yet to be used in thermal imaging for underground water leakage detection. Similarly, research regarding the use of geophones has yet to gain that much attention among researchers. Generally, leakage surveys are done in a larger area at a time. Automatic image processing and sorting methods are needed to process massive data. The researchers have used the manual method to differentiate the leakage. Some researchers have used simple algorithms, but they still need to be fully automated. The number of scenarios the researchers have considered could be much higher. All the researchers used a one-dimensional approach that used only IR images. Most researchers have taken a single shot of the location apart from Bach and Kodikara (2017); they have taken multiple shots of the same area and considered the temperature gradient. There is a potential to use ML to differentiate the scenario and conditions. It is essential to point out that the researchers have yet to use a combination of image processing and sound location to detect underground leakage.

4. Methodology

The proposed leakage detection method was a two-step approach. The two-step method ensured speed and accuracy. In the first step, thermal images were used to survey a large area in a short time, and this ensured the speed of leakage surveying. In the second step, a stereo geophone was used to verify the leakage using audio after the first stage of surveying. It also provided the accurate location of the underground leakage.

The methodology of the research is shown in Fig. 3. Step one is based on the thermal signature of the soil. Underground water leakage would change the thermal property of the nearby soil as water has a higher thermal mass than dry soil. Step two was based on the sound emitted from the leakage location. Water leakage would also generate noise underground. In this research, we used the thermal signature to locate the water leakage location and the leakage audio to verify the first step and pinpoint the underground leakage point.

The step one was based on thermal images. We could see from Table 1 that there was a lack of a thermal anomaly dataset for water leakage detection, so we conducted an experiment to generate the data.

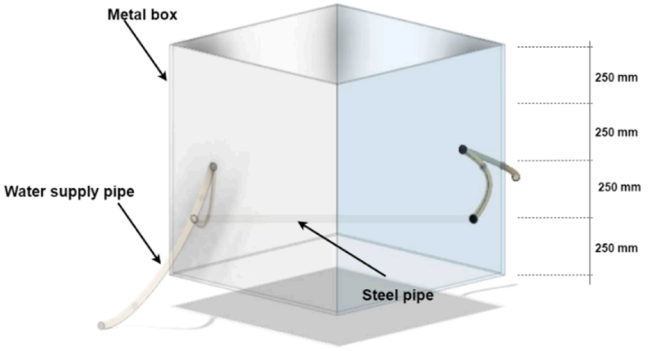


Fig. 4. 3D model of the testbed.

We set up a 1m3 metal box to create different underground leak conditions. The soil height above the pipe was changed from 250 mm to 750 mm in 250 mm steps. The images were captured from 7am to 7pm with fifteen minutes of intervals to capture the different conditions of leakage signature. The variation of the thermal signature was captured with a thermal camera. The images were categorized into two categories: "leak" and "non-leak" images. The data was formatted and labelled, and a dataset was made. The training dataset was rotated, flipped, translated and scaled to augment the dataset. Region of Interest (ROI) was separated via image segmentation. Image segmentation was done with Otsu thresholding, cropping, and morphological opening. Then, Non-Local Means Denoising (NLMD) was applied to reduce the noise. Then, the image was enhanced with contrast-limited adaptive histogram equalization (CLAHE), and finally, the ROI was used to train the ML model. The model was a data-driven hybrid model consisting of convolutional neural networks (CNN), long short-term memory (LSTM) and soft attention layers. The ML model was fine-tuned until it performed with more than 90 % accuracy. Then, an ablation study was done to tune the model further. This model would show us the approximate location of the leak.

In the second stage, we used a hydrophone to more accurately verify and localize the leakage. First, we used the CAMIL[1] dataset to make an ML model for sound source localization and study its performance. The interaural time difference (ITD) and the interaural phase difference (IPD) were calculated for each frequency of the audio as features. We fine-tuned the ML model until its accuracy was more than 90 %. This model was also a data-driven one-dimensional CNN model that used ITD and IPD features. This model was used as a base model for transfer learning in the next step. We set up an experiment and used geophones with a buried pipe to capture the audio and record the angular location of the leakage. We varied water pressure between 30 and 60 psi while we recorded the audio. We formatted, cleaned, and labelled the audio to make the dataset. We also used ITD and IPD features from our dataset and used transfer learning to develop the final model.

5. Underground water leak detection using thermal images

This section discusses the experiments we conducted and the ML algorithms development process for leakage detection using thermal images.

5.1. Experiment

The experiment was conducted in Casuarina, Darwin. Darwin soil mainly comprised Kandosol and Hydrosol along the coastal region and Tenosol and Sodosol in the inland (Shi et al., 2023). Kandosol and Hydrosol soils had a gravel range of 2 % −50% and a clay range of 10 % - 35

[1] Online at: https://team.inria.fr/perception/the-camil-dataset/

(a) (b)

Fig. 5. The container and the pipe setup. (a) shows the whole testbed and (b) shows the pipe setup before burying.

Fig. 6. FLIR thermal camera.

%; the remaining portion of soil was organic matter and sand (RF, 2002). As most of the cities in Darwin were along the coastal belt, Kandosol and Hydrosol soil were chosen for the experiment.

5.2. Test bed preparation

In Figs. 4, a 3D rendering of the testbed is shown. The testbed was made of steel sheets. An iron pipe rested on the 250 mm soil bed. A set of flexible pipes was used to supply and drain the water. Generally, in Australia, the supply pipe is buried 0.75 m below the ground (Territory, 2020). In the garden and the backyard, the pipe is buried around 0.5 m (Territory, 2020). To capture this range of buried pipes, a 1 m high, 1 m width and 1 m deep testbed was made.

Generally, PVC and ductile iron pipes were used in the drinking water system (Australia, 2011). The water supply hierarchy had five types of pipelines. The 'Bulk trunk main' (diameter: 250 mm upwards) collected water from sources, and the 'Bulk trunk' (200 mm to 250 mm) supplied water to service reservoirs. The 'Principal main' (100 mm to 200 mm) delivered water to suburbs and the 'Rider main' (diameter: 50 mm) supplied suburban areas. The fifth stage included council service pipes, private supply pipes, and connection pipes, all with a diameter of 20 mm (Code, 2011). For this reason, 20 mm diameter ductile iron pipe was used for the experiment.

The experimental setup is shown in Fig. 5. Fig. 5(a) showed the $1m^3$ metal container. Fig. 5(b) showed the pipe setup. The container had holes in the wall for running pipe through it. The pipe was resting on top of 250 mm of soil. We gradually filled the container with soil as we kept taking thermal photographs of the soil at different times of the day.

5.3. Data acquisition

IR cameras were expensive in the nineties, costing around ten thousand dollars. Recently, with the advent of mobile technology, an IR camera could be bought for under 450 AUD. FLIR ONE Pro LT Thermal Imaging Camera was an example of such a camera. This opened a new opportunity to explore. The thermal sensitivity of the camera was 100mK, and it could operate between 0 °C to 35 °C. It had a fixed focal length of 15 cm to infinity. The camera used multispectral dynamic imaging techniques to add visible light to thermal images (FLIR, 2024). This addition of visible light to the thermal image added clarity, edge, and outline.

Fig. 6 shows an image of the camera. It could connect to any mobile device that has a USB-C connection. As an image-based approach to detect leakage did not require specialized equipment, it would be fast and cheap to survey a large area. Details of the proposed data acquisition methodology are given below.

Table 2
Dataset summary.

Data type	Description
Leak size	3mm
Soil depth	250 mm, 500 mm, and 750 mm
Image capture time	7am to 6pm
Image capture interval	15 min
Number of leak images	4254
Number of non-leak images	1374
Total number of images	5628

5.4. Dataset

A summary of the dataset is provided in Table 2.

The diameter of the leak was 3 mm, and the soil depths were 250 mm, 500 mm and 750 mm receptively. For each soil depth setup, we conducted the experiment for a week. The IR and RGB photos were taken fifteen minutes apart from 7am to 6pm. Each time, a minimum of three RGB and three IR images were taken. Fig. 7 showed the RGB and IR images side by side. Fig. 7 (a-1, a-2, b-1, and b-2) showed the "non-leak" RGB and IR images. In the case of Fig. 7 (a-1, a-2), the IR image showed a uniform heat distribution in the morning, and there was no heat patch. In the case of Fig. 7 (b-1, and b-2), shadows were on the corner, and we could see a thermal patch there. Fig. 7 (c-1, c-2, D-1, and D-2) showed images of "leak" conditions. In the case of Fig. 7 (c-1, and c-2,), the leakage wet patch was visible in both IR and RGB images. However, Fig. 7 (d-1, and D-2) showed that it was not evident from the RGB image that there was wet soil underneath the dry topsoil, but with the IR image, the wet patch underneath was visible. We also placed dried leaves, grass, and plastic derbies on the surface while capturing the thermal images to simulate real-life surface conditions.

A dataset was made from the captured images. The dataset had predominately two classes. The first one would include the image of a leaked surface, and the second one would include the images of a non-leaked surface. The dataset would include the soil's temperature gradient colour and the leaked surface's IR signature. There was a total of 4254 of leak images and 1374 of non-leak images. Fig. 8 shows the thermographic images of leak conditions. We could see underground water's effect on soil's thermal characteristics in Fig. 8. A sharp patch of cold region was visible on the leakage images. On the other hand, Fig. 9

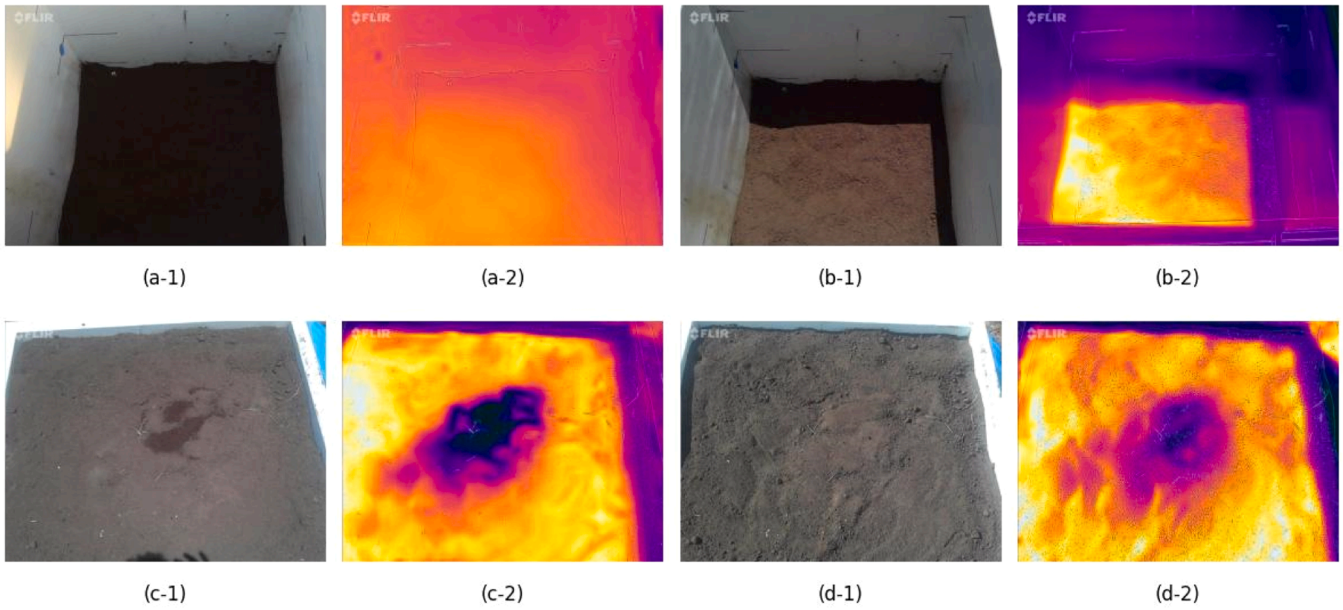


Fig. 7. Snapshot of the dataset.

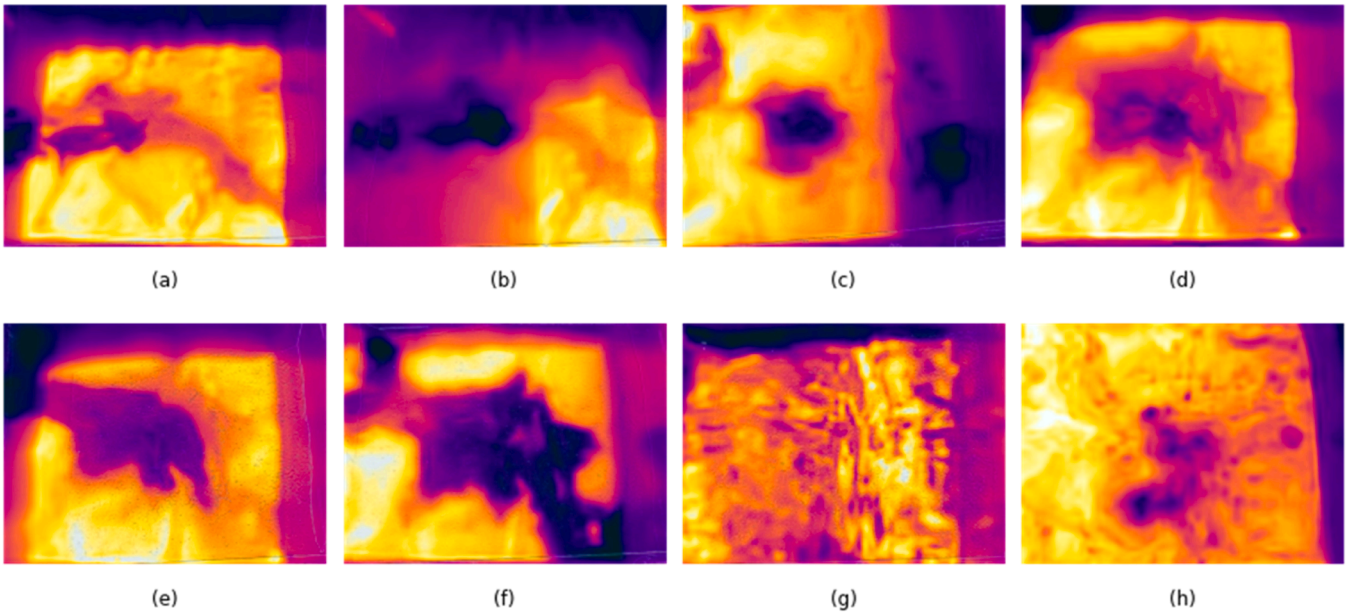


Fig. 8. Leak images.

shows the thermographic images of non-leak conditions in the dataset. A gradual and linear temperature gradient was present in the non-leak images, and it was due to shades of the surrounding structures.

5.5. Data augmentation

Image augmentation techniques are used in machine learning to expand the size and diversity of the training data. This diversity is obtained by applying various transformations to the dataset. Image augmentation produces images similar to the dataset photos but adds the differences frequently seen in real-world situations. Fig. 10(a) shows the original image, and Fig. 10 (b, c, d, and e) shows the augmented images. Fig. 10(b) shows the random rotation. The rotation range is from -90° to $+90^{\circ}$ with 30° steps. Eq. (1) shows the transformed coordinate. It is presented below:

$$\begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{1}$$

Here, f_x and f_y represent the transformed coordinate values, and x and y refer to the original image's pixel value. When an image is rotated, empty space appears at the edges. This space can be filled in a few ways. The first one is the constant fill, where empty areas are filled with a single value (black = 0, white = 255). The second one is to fill the empty areas with noise. The noise can be random or Gaussian noise. The padding keeps the image the same size after the shifting process. Padding with a constant value (c) during this process would involve replacing empty pixels. We filled the empty spaces with zero (black) values.

Fig. 10(c) shows the horizontal flip. Flipping the image across to its horizontal axis is known as horizontal flipping. In horizontal flipping,

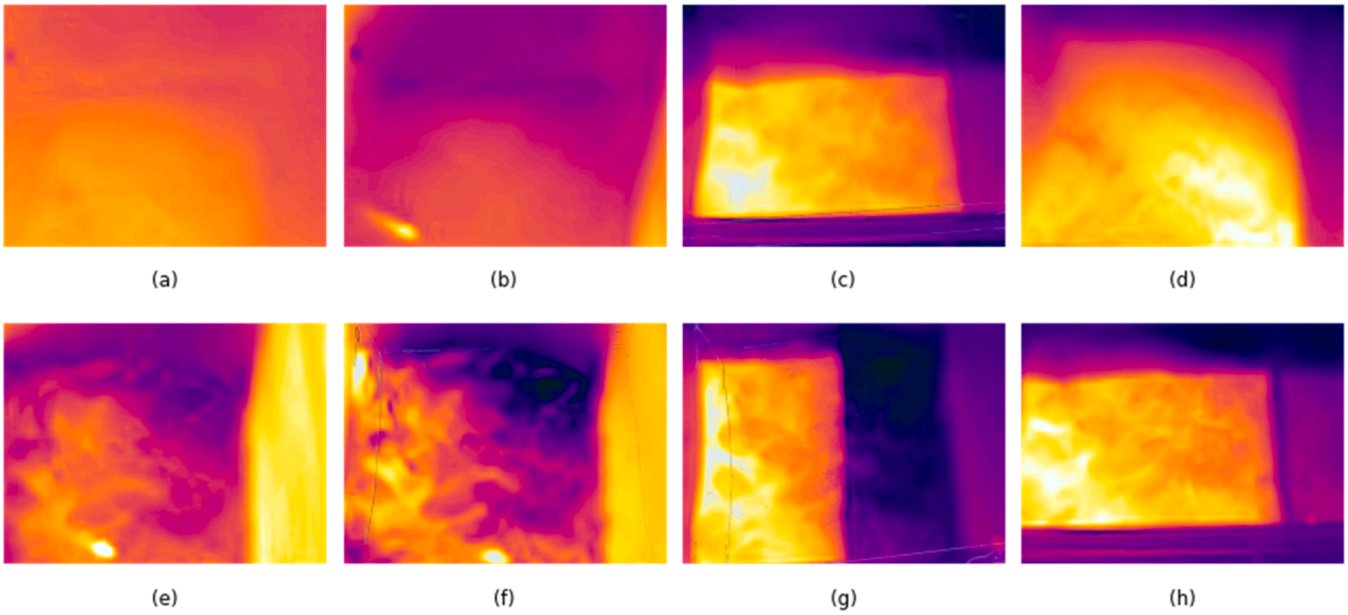


Fig. 9. Non-leak images.

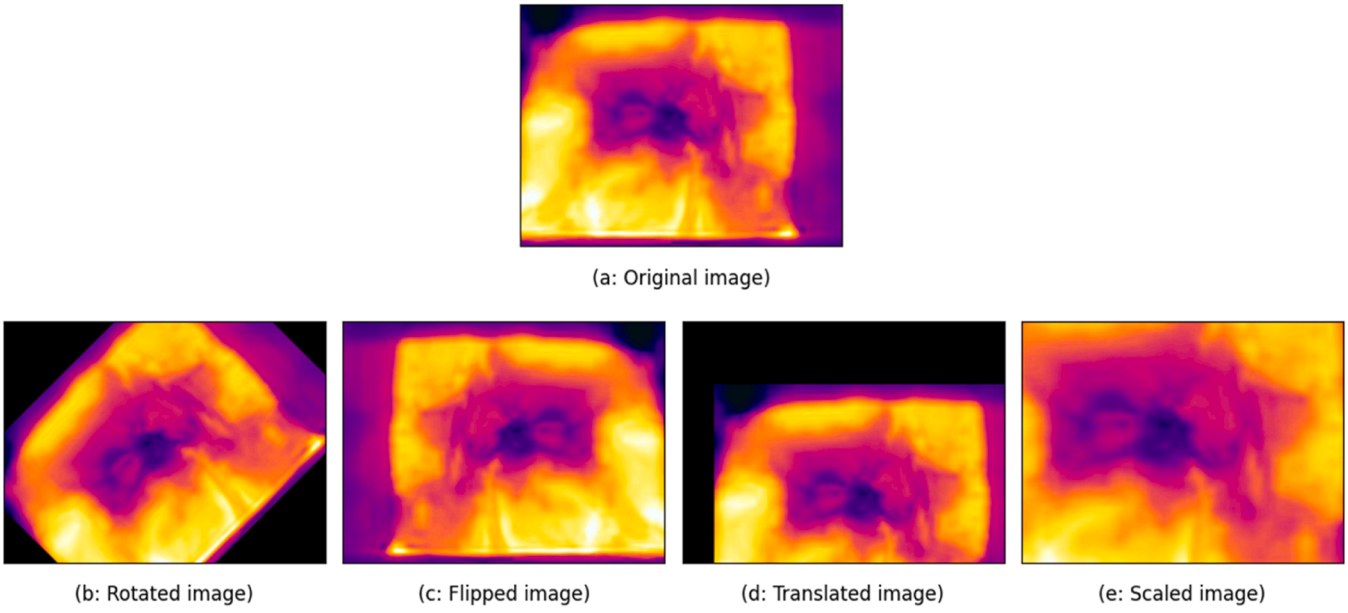


Fig. 10. Data augmentation.

the left sides are flipped to the right side and vice versa. The Eq. (2) shows the equation of the transformed coordinate. It is presented below:

$$\begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{2}$$

Here f_x and f_y represent the transformed coordinate values, and x and y refer to the original image's pixel value.

Fig. 10(d) shows the translation of the image. Moving images sideways or up and down (shifting) helps prevent the model from relying on specific object positions in the data (Shorten & Khoshgoftaar, 2019). Similar to the rotation, shifting an image creates space, and we fill it with zero value.

Mathematically, it can be written as,

$$f'(x',y') = f(x - T_x, y - T_y) \tag{3}$$

In Eq. (3), $f'(x',y')$ represents the transformed coordinate values and, x and y refer to the original image's pixel value. T_x and T_y define the amount of horizontal and vertical movement, respectively. Padding with a constant value (c) during this process would involve replacing empty pixels.

Fig. 10(e) shows the scaling. Scaling images by zooming in or out is a common augmentation technique. It helps the model learn features at different scales. Zooming in or out can introduce empty space around the image. Similar to rotation, this space can be filled with padding discussed in the rotation technique.

Mathematically, it can be written as,

$$f'(x',y') = f(S * x, S * y) \tag{4}$$

In Eq. (4), $f'(x',y')$ represents the new coordinates after zooming. S is the scaling factor. When S is greater than 1, it represents zooming in

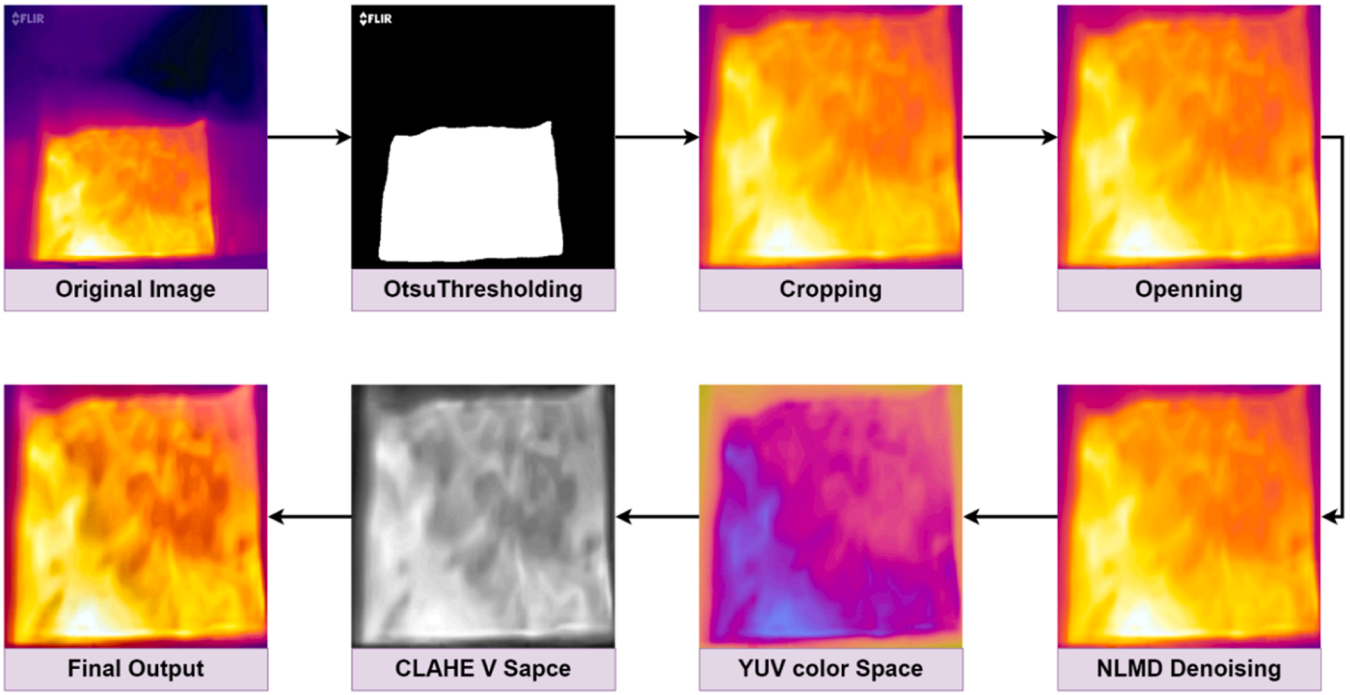


Fig. 11. Image preprocessing sequence.

(magnification). Conversely, S between 0 and 1 signifies zooming out. Padding with a constant value (c) during zooming would involve filling the empty space with c.

Augmentation helps generalize and make the model robust. This approach helped to reduce overfitting and helped the model extract meaningful features from the dataset. We applied these augmentations during the model training.

5.6. Image preprocessing

OpenCV v4.8.0 was used for image processing. A sample of the image preprocessing source code is available at the GitHub repository.[2] The image processing sequencing is presented in Fig. 11. The first image of Fig. 11 was the original image from the dataset, and we could see the difference in sunlight contact on the soil, which made three distinctive patterns. The bright orange portion was where there was direct sunlight for an extended period, and the soil was warm; the dark portion on the top was where there was no sunlight and was under shadow. The other portions had indirect sunlight and were purple.

We were interested in the bright orange portion and used Otsu's thresholding to separate the background from the interest zone. Eqs. (5) and (6) show the thresholding functions.

σ_B^2 are functions of the threshold at level k,

$$\sigma_B^2\ (k) = \frac{[\mu_T \omega(k) - \mu(k)]^2}{\omega(k)[1-\omega(k)]} \tag{5}$$

Here, μ_T represents the mean intensity and $\mu(k)$ represents the mean intensity at level k. The weight of the classes is represented by $\omega(k)$.

The optimal threshold at level $k^*$is,

$$\sigma_B^2(k^*) = max_{1\le k<L}\sigma_B^2(k) \tag{6}$$

Here, L represents the grey levels of the pixels. Otsu's thresholding was commonly used to determine the optimal threshold value to separate the foreground and background. It maximized the variance between the two classes. The algorithm converts an image to a binary image. If the intensity of the pixel value is greater than the Otsu threshold value, then the pixel is turned to black; if the intensity is less than or equal to the threshold, then the output pixel is white. The second image was after the application of Otsu thresholding (Otsu, 1979).

In the third stage, we used the original image and the Otsu threshold to crop the region of interest.

In the fourth stage, we used morphological opening. The morphological opening involves two operations: erosion and dilation. It was done to reduce noise and smooth out the contours. The (7) shows the mathematical formula of morphological opening.

$$(A,B) = (A\ \ominus B)\ \oplus B \tag{7}$$

Where, the morphological opening of an image A by a structuring element B is performed by applying erosion $\ominus$ operation followed by dilation operation $\oplus$.

In the fifth state, we applied Non-Local Means Denoising (NLMD). NLMD is a type of filter that operates by averaging the pixel value in a neighborhood around a target pixel. NLMD incorporated non-local information, which meant it considered similar patches from different regions of the image (Buades et al., 2011). For denoising a color image $u = (u_1, u_2, u_3)$ from pixel p to q, the Eqs. (7) and (8) are executed.

$$\hat{u}_i(p) = \frac{1}{C(p)} \sum_{q\epsilon B(p,r)} u_i(q)\ \omega(p,q) \tag{7a}$$

$$C(p) = \sum_{q\epsilon B(p,r)} \omega(p,q) \tag{8}$$

Where i can be written as $i = 1,\ 2,\ 3$ and $B(p,r)$ indicates a neighborhood centered at pixel p and with size radius of r pixels.

The weight ω can be determined by the squared Frobenius norm distance. The Frobenius norm distance decay between color patches with centers at p and q, which decay under a Gaussian kernel (p,q).

In the sixth stage, we converted the color space to YUV color space and used contrast-limited adaptive histogram equalization (CLAHE) in the Y channel to enhance the image (Reza, 2004). The final output is shown in the last image.

[2] Online at: https://github.com/shuzonudas/Leak-localization-ML-models.

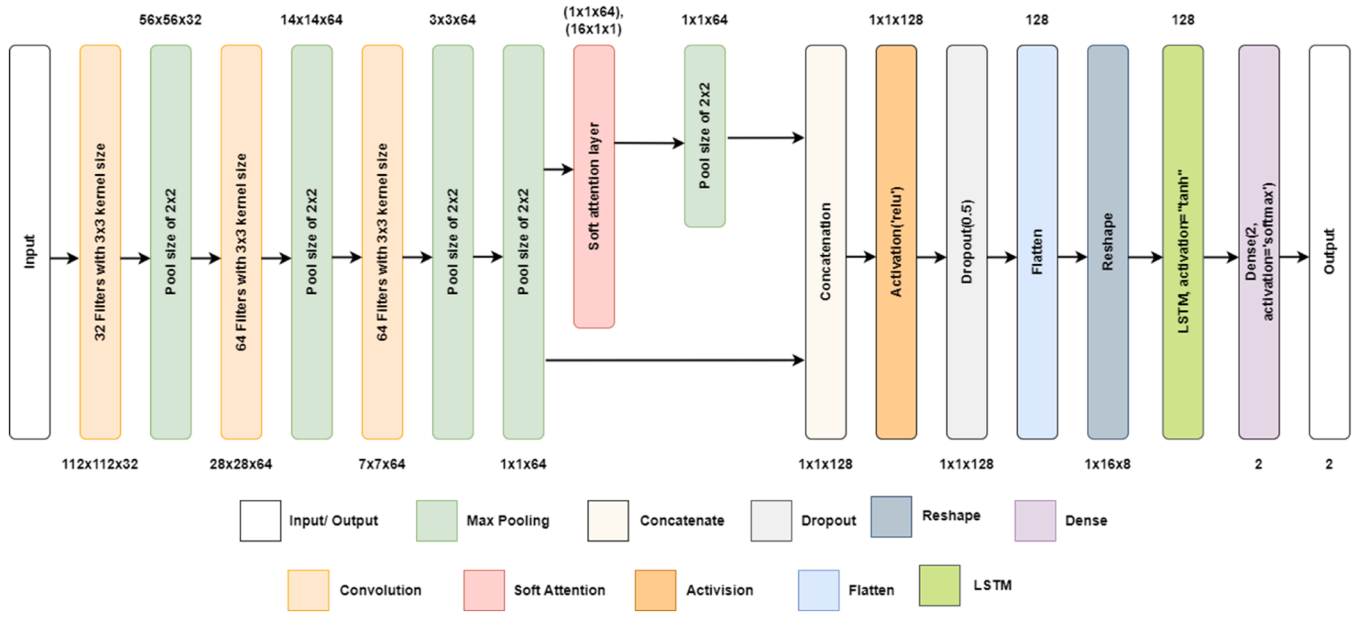


Fig. 12. Architecture of the proposed model.

5.7. Pretrained ML-models

This study used Xception, DenseNet-121, MobileNetV2, Inception-ResNetV2, and EffecientNetB0 pre-trained models. The dataset was divided into 70 % for training, 20 % for validation and 10 % for testing (Muraina, 2022). This ratio was used to train the model to compare the performance of the models. Only training data was augmented. The validation and the test data were kept as it were.

5.7.1. Xception

Xception stood for "Extreme Inception," and it was based on the idea of depth-wise separable convolutions. It was a variant of the Inception architecture. François Chollet introduced this architecture in 2016 (Chollet, 2017).

In Xception, the first step was the independent depth-wise convolution on each input channel. This step captured spatial dependencies by applying a small filter to each channel individually. The pointwise convolution was the second step that involved a 1×1 convolution to combine the output channels from the previous step. This operation helped capture channel-wise dependencies. By separating spatial and channel-wise operations, the model reduced the computational complexity, allowed for more efficient use of parameters and improved the model's ability to learn intricate features in the input data.

5.7.2. DenseNet121

Huang et al. introduced DenseNet-121 in 2016 (Huang et al., 2017). DenseNet was a densely connected convolutional network. The architecture of DenseNet-121 consisted of multiple dense blocks. In a dense block, each layer received input from all preceding layers and passed its output to all subsequent layers. This dense connectivity facilitated the flow of information throughout the network, promoting feature reuse and gradient propagation.

The feature maps from previous layers were combined with the channel dimension in each dense block. This created a single feature map containing information from all the layers before it. The dense concatenation operation ensured that each layer could access the combined knowledge of the entire network. This helped the model learn complete and strong representations. There were also transition layers that controlled the growth of feature maps. These transition layers employed 1×1 convolutions and average pooling to compress feature maps and down-sample the spatial dimensions.

The DenseNet-121 architecture enabled efficient parameter usage compared to traditional convolutional neural networks. The architecture also alleviates the vanishing gradient problem and enhances the model's gradient flow, leading to improved training and better overall performance.

5.7.3. MobileNetV2

The MobileNetv2 model was a deep-learning architecture introduced by Sandler et al. (2018). The model was specifically designed to achieve high accuracy while minimizing computational resources for mobile and embedded vision applications.

MobileNetv2 used inverted residual blocks consisting of a lightweight 1×1 convolution, a depth-wise separable convolution, and another 1×1 convolution to expand the channels. This structure allowed for efficient feature extraction and preserved representational power while minimizing computational cost. It also utilized linear bottlenecks with shortcut connections to improve information flow and gradient propagation. The linear bottlenecks reduced the dimensionality of the feature maps before the expensive depth-wise separable convolution, reducing computational requirements.

The architecture also included a width and resolution multiplier, providing control over the model's complexity and trade-off between accuracy and efficiency. Adjusting these multipliers allowed the model to be tailored to specific deployment scenarios with varying computational constraints.

5.7.4. EffecientNetB0

The EfficientNetB0 model was a deep learning architecture introduced by Tan and Le (2019). The architecture of EfficientNetB0 was based on a compound scaling method that uniformly scales the depth, width, and resolution of the network. This scaling strategy ensured an optimal balance between model complexity and resource efficiency.

EfficientNetB0 consisted of repeated blocks. The blocks contained a combination of depth-wise separable convolutions and squeeze-and-excitation operations. The depth-wise separable convolutions performed separate convolutions on each channel, followed by a pointwise convolution. This reduced computational complexity while capturing both spatial and channel-wise dependencies.

The squeeze-and-excitation operations further enhanced the model's representation power by adaptively recalibrating channel-wise feature responses. This allowed the model to focus on more informative features

Table 3
Architecture of the proposed model.

Layer Type	Output Shape	Parameters
input_1 (InputLayer)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 112, 112, 32)	896
max_pooling2d (MaxPooling2D)	(None, 56, 56, 32)	0
conv2d_1 (Conv2D)	(None, 28, 28, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_2 (Conv2D)	(None, 7, 7, 64)	36,928
max_pooling2d_2 (MaxPooling2D)	(None, 3, 3, 64)	0
max_pooling2d_3 (MaxPooling2D)	(None, 1, 1, 64)	0
soft_attention (SoftAttention)	[(None, 1, 1, 64), (None, 16, 1, 1)]	9232
max_pooling2d_5 (MaxPooling2D)	(None, 1, 1, 64)	0
max_pooling2d_4 (MaxPooling2D)	(None, 1, 1, 64)	0
Concatenate (Concatenate)	(None, 1, 1, 128)	0
Activation)	(None, 1, 1, 128)	0
Dropout (Dropout)	(None, 1, 1, 128)	0
Flatten (Flatten)	(None, 128)	0
Reshape (Reshape)	(None, 16, 8)	0
lstm (LSTM)	(None, 128)	70,144
dense (Dense)	(None, 2)	258

and improve discrimination.

5.8. Proposed model

In this study, we proposed a hybrid model. The model consists of convolution with soft attention layer and long short-term memory (LSTM) layers (Zhang et al., 2023). We used Python v3.10.12 as the programming language. NumPy v1.25.2, Pandas v2.0.3, Librosa v0.10.1, Scikit-learn v1.2.2, OpenCVv4.8.0 and TensorFlow v2.15.0 libraries were used for data manipulation and model development. We trained the model on Google Colab version 0.0.1a2. The ML model code[3] is on GitHub. The architecture of the model is described in the next section.

5.8.1. Model architecture

The model was designed for binary classification and was constructed sequentially. The data also flowed serially through the layers as the four types of layers were stacked continuously. The overview of the proposed model is presented in Fig. 12.

The proposed model was structured in sequence. There were convolution, soft attention, LSTM, and dense layers. The model took an input of three channel thermal images of 224×224 pixel size and generated a probability output.

The first was a Conv2D layer with a 3 × 3 kernel size and 32 filters. The activation function used in this layer was a rectified linear unit (ReLU). The layer took direct input from the input layer. The next layer was the MaxPooling2D layer with a pool size of 2 × 2, which performed spatial down sampling and dimensionality reduction. Two more sets of Conv2D and MaxPooling2D layers were added to extract hierarchical features from the input. In both sets of layers, the filter size was 64, and the pool size was 2 × 2. A second set of MaxPooling2D layers was added here to reduce dimensionality and improve model efficiency.

A soft attention layer helped the model to focus on the relevant area of thermal images and pass the refined information to the MaxPooling layer. The outputs of the convolution layer and the soft attention were concatenated and passed through the ReLU activation function, followed by a dropout regularization.

The flattened output was then reshaped and fed into a Long Short-Term Memory (LSTM) layer with 128 units and a tanh activation function. LSTM was used here to utilize the temporal anomaly of the dataset. Finally, a dense layer with two units and a softmax activation function was employed for binary classification.

Table 4
Ablation study summary table.

Configuration No	Soft Attention Layer	Accuracy (%)	Finding
Ablation Study 1: Altering the Soft Attention layer			
1	**1**	**93.79**	**Highest accuracy**
2	2	93.43	Accuracy dropped
Ablation Study 2: Altering the LSTM Block			
Configuration No	LSTM Block	Accuracy (%)	Finding
1	**1**	**93.79**	**Previous accuracy**
2	2	93.26	Accuracy dropped
Ablation Study 3: Altering the Batch Size			
Configuration No	Batch Size	Accuracy (%)	Finding
1	16	88.47	Accuracy dropped
2	**32**	**93.97**	**Highest accuracy**
3	64	93.79	Previous accuracy
4	128	93.61	Accuracy dropped
Ablation Study 4: Altering the Dropout Rate			
Configuration No	Dropout rate	Accuracy (%)	Finding
1	0.1	93.61	Accuracy dropped
2	0.3	93.79	Accuracy dropped
3	**0.5**	**93.97**	**Previous accuracy**
4	0.7	93.62	Accuracy dropped
Ablation Study 5: Altering the Optimizer			
Configuration No	Optimizer	Accuracy (%)	Finding
1	Adam	93.97	Previous accuracy
2	Adamax	92.37	Accuracy dropped
3	RMSprop	87.06	Accuracy dropped
4	**Nadam**	**95.20**	**Highest accuracy**
Ablation study 6: Altering the Learning rate			
Configuration No	Learning rate	Accuracy (%)	Finding
2	0.001	89.53	Accuracy dropped
3	**0.0001**	**95.20**	**Previous accuracy**
4	0.00001	94.50	Accuracy dropped

The detailed architecture of the proposed model is given in Table 3. The model took input using the input layer. The input was a 224×224 pixel, three-channel thermal image. There are three sets of Conv2D and MaxPooling2D layers. The first set has 32 feature maps, and the latter two sets have 64 feature maps, respectively. These layers hold the feature maps and, at the same time, reduce the spatial dimensionality. These sets have 896, 18,496 and 36,928 parameters, respectively. The network then uses another Maxpooling layer and feeds the information to a SoftAttention layer for focused attention on relevant thermal features, and it has 9232 parameters.

The context vector outputs from the SoftAttention layer were then passed through one MaxPool layers, and a concatenation layer was used to combine them with the convolution layer output. The combined output was then flattened, reshaped, and fed through the LSTM layer. The LSTM layer had 70,144 parameters. Finally, a dense layer with two outputs was added, and these two outputs were used for binary classification. The dense layer had 258 parameters. The model had a total of 1,35,954 trainable parameters.

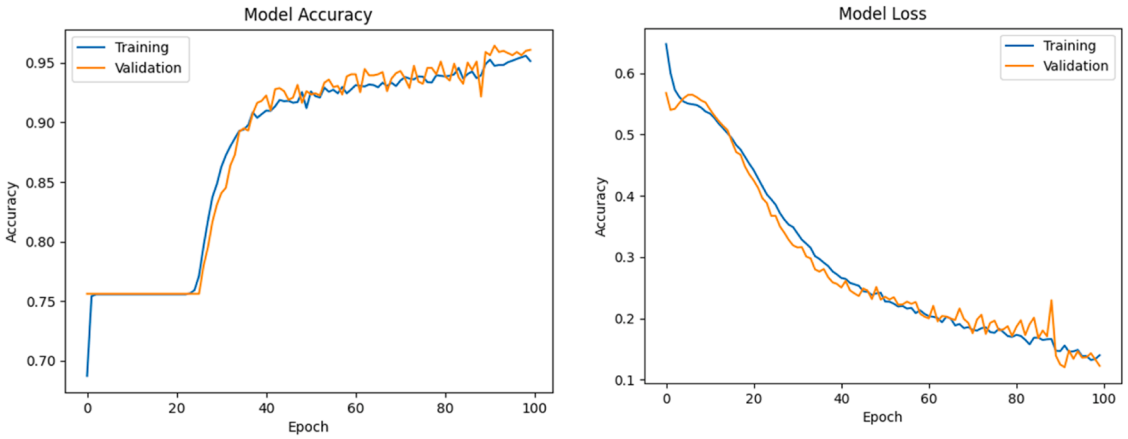
5.8.2. Performance comparison

The confusion matrix was used to calculate the accuracy, recall, precision, specificity, and F1 score of the model (Shukla & Piratla, 2020). The following expressions (Eqs. (1) to 5) were used to calculate the mentioned parameters,

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1a}$$

$$Precision = \frac{TP}{TP + FP} \tag{2a}$$

[3] Online at: https://github.com/shuzonudas/Leak-localization-ML-models.



(a) Model training and validation accuracy. (b) Model training and validationo loss.

Fig. 13. Training and validation graph of accuracy and loss.

$$Sensitivity = \frac{TP}{TP + FN} \tag{3a}$$

$$Specificity = \frac{TN}{TN + FP} \tag{4a}$$

$$F1 - Score = 2\left(\frac{Precision \times Sensitivity}{Precision + Sensitivity}\right) \tag{5a}$$

Here, True Positive (TP) meant that the model had detected a leakage, and there was a leakage. True Negative (TN) meant that the model had detected a non-leakage, and there was also no leak. False Positive (FP) meant the ML model had detected a leakage, but none exists. Finally, False Negative (FN) meant the ML model had detected a non-leak condition, but there was leakage on the ground. The confusion matrix was used to determine TP, TN, FP, and FN values.

5.8.3. Ablation study

In this part of the paper, we studied the impact of various configurations and components of the proposed model. We systematically altered different aspects of the model architecture, including the soft attention layer, LSTM block, batch size, dropout rate, loss function, optimizer, and learning rate. We could gain insights into their contributions to the model's overall accuracy by systematically removing or modifying these components. A summary of the ablation study is presented in Table 4.

The base model we used here had one SoftAttention layer and one LSTM layer. The batch size was 64, dropout rate was 0.5, with Adam optimizer, and with the learning rate of 0.0001.

At first, we changed the soft attention layer. Adding only one attention layer increased the overall model accuracy but adding the second attention layer decreased accuracy. A similar trend was observed with the LSTM layer as well; adding the second layer decreased the overall model accuracy. The optimal batch size was 32 for training this model efficiently. Increasing or decreasing the batch size beyond 32 resulted in decreased accuracy. The highest accuracy was obtained with a dropout rate of 0.5. Dropout was essential for improving the model's generalization ability and helped prevent overfitting of the model. The combination of the Nadam optimizer and a learning rate of 0.001 produced the model with the best performance. The evaluation metrics for the best-performing ML model are: accuracy of 0.9520, precision of 0.9835, sensitivity of 0.9543, specificity of 0.9435, and an F1-score value of 0.9687.

5.8.4. Model performance

A hybrid architecture model was used in this experiment. Fig. 13 shows the accuracy and loss curve of the training and validation dataset. The accuracy measured the model's performance and was expressed as a percentage in the curve. The model started with zero accuracy and stalled at around 75 % till 30 epochs. The model then further improved with the number of epochs proceeds. The training and validation accuracy curves were quite similar, but we could see a minor oscillation in the validation curve. Then the model accuracy steadily increased to 94.34 % with training dataset and 95.02 % with validation dataset

Both the training and validation loss curves also look very similar. The loss consistently improved as epochs progressed. Similar to the accuracy curve, there was a minor oscillation in the validation curve, but the trend was similar. The training loss and validation losses were 0.1589 and 0.1632 respectively.

The confusion matrix is presented in Fig. 14. It showed that out of 562 testing images, our model could correctly classify 535 images. This meant our model's accuracy was 95.20 %. The rest of the parameters regarding the model's performance are discussed in the next section.

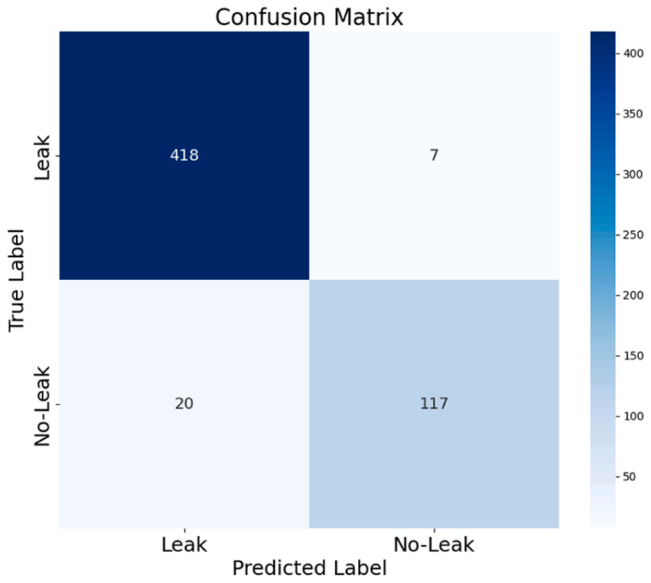


Fig. 14. Confusion matrix.

5.9. Comparative analysis

In this section, we compared our work with popular pre-trained

Table 5
Model comparison summary.

Model	Accuracy	Precision	Sensitivity	Specificity	F1-score
Xception	0. 9060	0.8941	0.9793	0.7443	0.9348
DenseNet121	0. 8280	0.84	0.9249	0.618	0.8804
MobileNetV2	0.9167	0.9247	0.9632	0.7949	0.9435
EfficientNetB0	0.7713	0.7859	0.8978	0.5260	0.8381
Proposed model	**0.9520**	**0.9835**	**0. 9543**	**0.9435**	**0.9687**

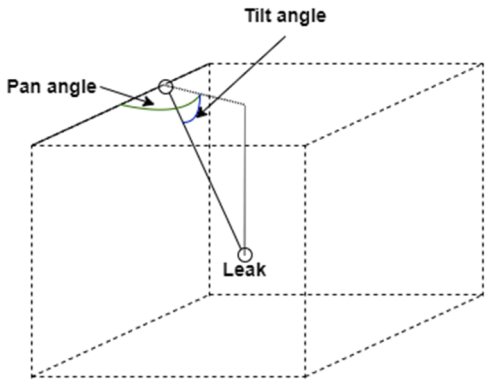


Fig. 15. Leakage sound localization.

models. These models included DenseNet121, EfficientNetB0, InceptionResNetV2, MobileNetV2, and Xception. The summary of the comparison is shown in Table 5.

The table shows accuracy, precision, sensitivity, specificity, and F-1 score. The accuracy metric measured how often the model correctly predicted the outcome for a given input. The above table showed that our proposed model had the highest accuracy of 0.9520. Precision measured the model's ability to identify positive cases, and specificity, on the other hand, measured the model's ability to identify negative cases correctly. Our proposed model had the precision and the specificity of 0.9835 and 0.9543. The sensitivity or recall value told the user the correct proportion of actual positives that the model correctly identified. Our proposed model had the highest sensitivity of 0.9543. F1-score, on the other hand, measured the harmonic mean of the model's precision and sensitivity and the proposed model also excelled compared to the other pre-trained models and achieved 0.9687.

6. Underground leak localization using geophones

Binaural audio cues could be used for sound source localization. There is a lot of research to localize audio sources in the natural environment (Liaquat et al., 2021). This section of the study focused on designing and developing an ML model for localizing the source of water leaks in underground settings. Fig. 15 shows the relationship between geophone locations and the leak location.

After locating the approximate location with thermal imaging, a pair of geophones would be used to get the leakage sound. The distances of the geophones to the audio source would be different. Due to the distance difference, the leakage signal would have time and phase differences. Desai et al. showed that the human auditory system model was popular in sound source localization, and common binaural cues were ITD and IPD. ITD is the difference in arrival time between a sound at the left and right ears. The average human head has a width of around 20–22 cm. If a sound comes from the left side of the head, it reaches the right ear after 0.6 ms compared to the left ear. IPD, on the other hand, refers to the phase difference of the audio signal. IPD can be calculated using the ITD and the audio signal frequency. Time-frequency analysis can be used to make a mathematical model of the human auditory

Fig. 16. Sensor on the experimental test bed.

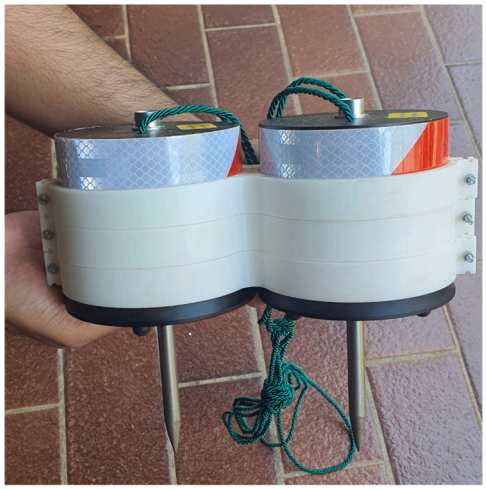


Fig. 17. The geophone sensor pair.

system using ITD and IPD (Desai & Mehendale, 2022). We would use the time and the phase difference information to develop the ML model. The position of the sound source could be classified as pan and tilt angle. The pan and tilt angle direction are shown in Fig. 15. The model developed would provide pan angles only, as pan angle would be easy to measure on the surface. Sampling the audio at a few proximate locations could yield precise location information on the ground.

6.1. Experimental setup

We used the same soil and the test configuration, as discussed in Section 5.1. We let the soil rest for three months so it could get compaction by its weight. The soil settled an average of 82 mm after the resting period. We marked the leakage location with a metal marker. Fig. 16 shows the sensor placed in the testbed. The vertical bar of stainless-steel metal indicates the known leak position as marked by a white box in the Fig. 16. We used a ruler and pan angle template to place the sensor on the ground at specific angles. A channel was made to keep the water from saturating the sensors.

6.2. Data acquisition

A pair of geophones were used to capture the leaked audio data. Fig. 17 shows the geophones in detail. 3D-printed brackets were used to hold the ceramic geophones in place. The metal rods on the bottom acted as probes and were inserted into the ground for audio recording.

A "Digitnow" stereo audio capture card was used to capture the data. The leakage location was kept stationary, and we changed the water pressure and the valve opening while capturing the audio so that we could have some variation in the dataset. We also placed the sensor on different places to increase the sample size. The pressure range was between 30 and 60 psi. Details of the dataset are presented in section "6.6.1. Generated Leak Audio Dataset."

Table 6
CAMIL dataset summary.

Data type		Descriptions
"Background noise" clips		10,800
"White noise" clips		10,800
Random utterances clips		10,800
Total number of audio clips		32,400
Range	Pan	−180 to +180 °
	Tilt	−60 to +60 °
Sample rate		48 kHz
Audio format		Wav
Audio channel		Stereo

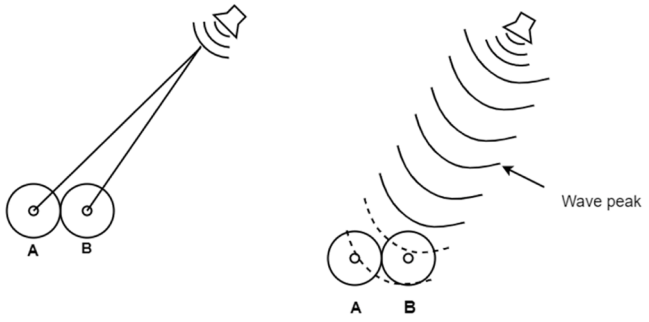


Fig. 18. ITD (left) and TPD (right) cues.

6.3. CAMIL dataset

Due to the limited training dataset, we adopted a two-stage transfer learning approach to develop the model (Shi et al., 2020; Shi et al., 2023). Initially, the CAMIL dataset was the foundation for creating a baseline ML model. Hyperparameters of the baseline ML model were fine-tuned to optimize model performance, and the best-performing model was chosen as the pre-trained model. Then, the "6.6.1. Generated Leak Audio Dataset" data was used to refine and develop the final model further using the pre-trained foundation. The details of the CAMIL dataset are presented in Table 6.

The CAMIL was generated to experiment with Computational Auditory Scene Analysis (DELEFORGE, 2012). It was designed to gather a very large number of binaural audio sounds with robot POPEYE—a total of 32,400 samples with pan and tilt angles. The data was recorded with a Sennheiser MKE 2002 dummy head, linked to a computer via a Behringer ADA8000 Ultragain Pro-8 digital external sound card. There were 10,800 motor states. The pan state covered angles from −180 to +180° and the tilt angle covered −60 to +60°. As the speaker produced various sounds, the robot captured one recording every 2°, and three separate static binaural recordings were collected each time. Recordings were conducted while a stationary loudspeaker emitted sounds with diverse properties, including random spectrum sounds, white noise, speech, and music. Each recording was annotated with the robot's corresponding ground truth motor coordinates.

Our experiment focused on a narrower pan angle range from −90 to +90° and all the pan angles within the field. We categorized the pan data into 18 classes with intervals of 10° each. Out of 10,800 clips from random utterances, the considered range covered 5005 clips from the dataset.

6.4. Feature extraction

We used two features to train the ML model: Interaural Phase Difference (IPD) and Interaural Time Difference (ITD). These two were essential cues for sound source localization. Fig. 18 shows the ITD and IPD cues for two sensors.

ITD and IPD were essential concepts in human auditory perception. It was crucial for our ability to localize sounds in space. Due to the separation of our ears, a sound emitting from a point would reach a slightly different time between two ears. The difference between the two arrival times provided crucial information about the sound source's location. On the left of Fig. 18, we could see that the travel time of the audio signals were different. Audio would reach sensor B first; then, it would reach sensor A. So, the time difference it took to reach two sensors was called ITD. On the other hand, IPD referred to the difference in the wave phase that reached each sensor. On the right of Fig. 18, we could see that wave peaks reached the sensor at slightly different times, thus resulting in phase differences. IPD and ITD were dependent on the frequency of the sound waves. We extracted ITD and IPD features from 200 Hz to 10,200 Hz in both datasets. This provided us with a total of twenty thousand data points, with ten thousand points dedicated to ITD and an additional ten thousand points allocated for IPD for each audio clip. We estimated the ITD based on the algorithm proposed by Chisaki et al. (2008).

- The first step was applying the Fast Fourier transform (FFT) to the left $l(n)$ and right $r(n)$ audio signals. FFT converted the audio signals from the time domain to the frequency domain.

  The spectra $L(k)$ and $R(k)$ are obtained by FFT of the input signals. Here, k is the frequency bin index.
- The next step was to find the bin in the FFT result that corresponded to the target frequency. This was done by multiplying the target frequency by the length of the FFT and then dividing by the sample rate. The result was then converted to an integer to get the index of the bin.
- The IPD carried the location of each frequency component and could be obtained through a cross spectrum $C_{lr}(k)$ as shown in (9) and defined as,

$$C_{lr}(k) = L(k)R(k)^*, \tag{9}$$

Where * denotes the complex conjugate. IPD, $\theta_{lr}(k)$ was calculated for each frequency component using the cross-spectrum $C_{lr}(k)$ as shown in (10) and defined as,

$$\theta_{lr}(k) = \tan^{-1}\left\{\frac{Im[C_{lr}(k)]}{Re[C_{lr}(k)]}\right\} \tag{10}$$

Here, *Im* stood for imaginary component and *Re* stood for real component of the complex conjugate.

- The phase difference was in radians, but the ITD needed to be in seconds. Therefore, the phase difference was divided by 2 * pi * the target frequency, to convert it to seconds.

6.5. Proposed ssl model

For leak source localization, we proposed a 1D-CNN model. The model consisted of convolution layers, batch normalization layers and dense layers. The architecture of the model is described in the next section. The dataset was divided into 70 % for training, 10 % for validation and 20 % for testing (Muraina, 2022). We used Librosa v0.10.1 for audio processing. The rest of the libraries are the same (as discussed in Section 5.8). A sample of the feature extraction and ML model code[4] is available on GitHub.

6.5.1. Model architecture

The model was a classification model with 18 output classes. The model had two main parts. The first part was "feature extraction" section

[4] Online at: https://github.com/shuzonudas/Leak-localization-ML-models.

Table 7
Architecture of the proposed model.

Layer Type	Output Shape	Parameters
input_1 (InputLayer)	(None, 20,000, 1)	0
conv1d (Conv1D)	(None, 19,998, 64)	256
batch_normalization (Batch Normalization)	(None, 19,998, 64)	256
max_pooling1d (MaxPooling1D)	(None, 9999, 64)	0
conv1d_1 (Conv1D)	(None, 9997, 128)	24,704
batch_normalization_1 (BatchNormalization)	(None, 9997, 128)	512
max_pooling1d_1 (MaxPooling1D)	(None, 4998, 128)	0
conv1d_2 (Conv1D)	(None, 4996, 256)	98,560
batch_normalization_2 (BatchNormalization)	(None, 4996, 256)	1024
max_pooling1d_2 (MaxPooling1D)	(None, 2498, 256)	0
flatten (Flatten)	(None, 639,488)	0
dense (Dense)	(None, 256)	8,18,54,592
dense_1 (Dense)	(None, 128)	16,512
batch_normalization_3 (BatchNormalization)	(None, 128)	512
dropout (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 50)	6450
dense_3 (Dense)	(None, 18)	918

Table 8
Ablation study summary table.

Ablation Study 1: Altering the convolution			
Configuration No	No. of conv. layer	Accuracy (%)	Finding
1	2	93.91	Starting point
2	**3**	**94.81**	**Highest accuracy**
3	4	92.21	Accuracy dropped
4	5	87.71	Accuracy dropped
Ablation Study 2: Altering the kernel size			
Configuration No	Kernel size	Accuracy (%)	Finding
1	2	93.11	Accuracy dropped
2	**3**	**94.81**	**Previous accuracy**
3	4	90.91	Accuracy dropped
4	5	90.21	Accuracy dropped
Ablation Study 3: Altering the Batch Size			
Configuration No	Batch Size	Accuracy (%)	Finding
1	16	91.61	Accuracy dropped
2	**32**	**94.81**	**Previous accuracy**
3	64	91.51	Accuracy dropped
4	128	93.51	Accuracy dropped
Ablation Study 4: Altering the Dropout Rate			
Configuration No	Dropout rate	Accuracy (%)	Finding
1	0.1	92.21	Accuracy dropped
2	0.3	91.11	Accuracy dropped
3	**0.5**	**94.81**	**Previous accuracy**
4	0.7	93.31	Accuracy dropped
Ablation Study 5: Altering the Optimizer			
Configuration No	Optimizer	Accuracy (%)	Finding
1	**Adam**	**94.81**	**Previous accuracy**
2	Adamax	94.00	Accuracy dropped
3	RMSprop	93.21	Accuracy dropped
4	Nadam	92.61	Accuracy dropped
Ablation study 6: Altering the Activation function			
Configuration No	Activation function	Accuracy (%)	Finding
1	PReLU	93.31	Accuracy dropped
2	**Relu**	**94.81**	**Previous accuracy**
3	Leaky ReLu	94.01	Accuracy dropped
4	ELU	92.41	Accuracy dropped

and the second part were "classification" section. The feature extraction part had three sets of convolutions, batch normalization and pooling layer and this part was connected to "classification" section via a flatten layer. The classification section had two sets of dense layers connected via a batch normalization and dropout layer. The model was a 1D CNN model.

The details of the model layers are given in Table 7. The input layer took both ITD and IPD as input at the same time. There were twenty thousand input points for the time and phase difference of the stereo signals at 200 Hz to 10,200 Hz frequencies. There were three sets convolution layer followed by batch normalization layer and max pooling layers. The first convolution layer had 64 feature maps and the latter two had 128 and 256 feature maps respectively. The convolution layers had 256, 24,704, and 98,560 parameters respectively. Batch normalization used here for improving the training speed and stability of the model. It scaled and shifted the activation to normalize the input of the layer to keep the mean of the layer close to zero and standard deviation close to one. Maxpooling layer was also used here to reduce the spatial dimension while keeping the important information.

Finally, we used a combination of flattening layer along with dense layers for classification. There were four dense layers. They had 8,18,54,592, 16,512, 6450 and 918 parameters respectively. The last dense layer had eighteen outputs for the final classification. The model had a total of 8,20,04,296 parameters including, 8,20,03,144 trainable parameters and 1152 non-trainable parameters.

6.5.2. Ablation study

Like the previous ML model, we have done an ablation study of the proposed sound source localization model as well. A total of six ablation studies were carried out using the CAMIL dataset to obtain the best configuration of the model. Systematic alterations were made to the different aspect of the model's architecture including number of convolution layers, kernel size, batch size, dropout rate, loss function, optimizer, and activation function. A summary of the ablation study is shown in the Table 8.

We started with a base model with two dense layers, kernel size of 3, dropout rate of 0.5, and ReLu as an activation function. We set Adam as the optimizer, and the batch size for training the model was 32.

In the first case, we changed the number of convolution layers and observed that peak accuracy was achieved with three layers at 94.81 %. Adding subsequent layers caused the accuracy to drop. In the next phase, we altered the kernel size and increasing or decreasing the kernel size caused a drop in accuracy. The change in batch size and dropout rate had a negligible impact on the overall model accuracy. A batch size other than 32 and a dropout rate of 0.5 decreased the accuracy. We tested four optimizers in this experiment: Adam, Adamax, RMSprop and Nadam, with a 0.001 learning rate. The Adam optimizer achieved the highest accuracy of 94.81 %. In the case of the activation function, the ReLu performed the best, reaching 94.81 % accuracy. The other functions we tested were PReLu, Leaky ReLu and ELU.

6.5.3. Model performance

The model presented was a sequential model. Fig. 13 shows the accuracy and loss curve of training and validation dataset. Both training and validation accuracy curve had similar shapes. Within 20 epochs, the mode's accuracy reached more than 90 % and hovered around that point. We could see at around 20 epochs and 50 epochs, there were two sharp drops of accuracy but there was not significant improvement of the model's overall accuracy after 20 epochs. The training and validation loss curves were quite similar in shape as well, but we could see a minor oscillation in the validation curve. The model achieved 99.88 % accuracy with training dataset and 92.61 % accuracy with validation dataset. The model had training and validation loss of 0.0043 and 0.3318 respectively.

The confusion matrix is presented in Fig. 14. It showed that out of 1001 testing images our model could correctly classify 949 images. It meant our model's accuracy was 94.81 %.

6.6. Transfer learning

Transfer learning is an efficient method of deep learning. It involves using a pre-trained network for a specific application. As the limited size

Table 9
Dataset summary.

Data type	Description
Water pressure	30 ~ 60 psi
Total number of audio clips	635
Bit rate	141 kbps
Sample rate	44.1 kHz
Audio format	Wav
Audio channel	Stereo

of our collected data was limited, we applied transfer learning to train our model. We used the model in Section 6.5.1 as our base model for transfer learning.

6.6.1. Generated leak audio dataset

The final model was developed with the captured underground sound explained in "6.1. Experimental Setup" and "6.2. Data Acquisition" section. The summary of the dataset is in Table 9.

There were a total 635 clips of annotated data. The clips were annotated with associated pan angles. Each clip had a different length, and we spliced them at one second's length. We changed the water pressure to simulate different leak conditions when capturing the audio clips. All the audio recordings were captured at a sampling rate of 44.1 kHz, with a bit rate of 141 kbps, and saved in WAV format. The audio clips were in stereo format. The final dataset had nine classes with 20-degree intervals.

Figs. 19-21

6.6.2. Transfer learning

For transfer learning we used the same data split as the base model. In the transfer learning process, we changed the output layer from 18 classes to 9 classes and did not change any other part of the model architecture. We kept the model layers unfrozen during the transfer learning process so that network could undergo fine tuning in the process. Fig. 22 shows the accuracy and the loss curve of the model. Approximately after 20 epochs, the model accuracy and loss curve stabilize. The model achieved 100 % accuracy with training dataset and 88.98 % accuracy with validation dataset. The model had training and validation loss of 0.0040 and 0.5313 respectively.

Fig. 23 shows the confusion matrix of the model and we could see

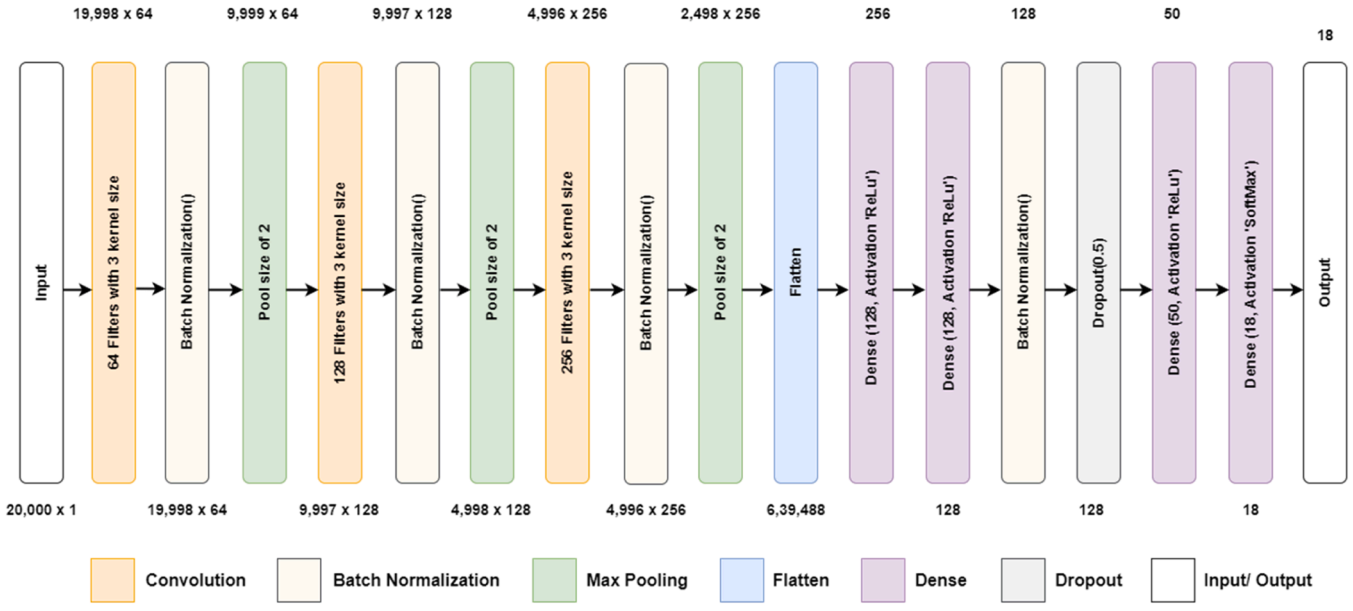
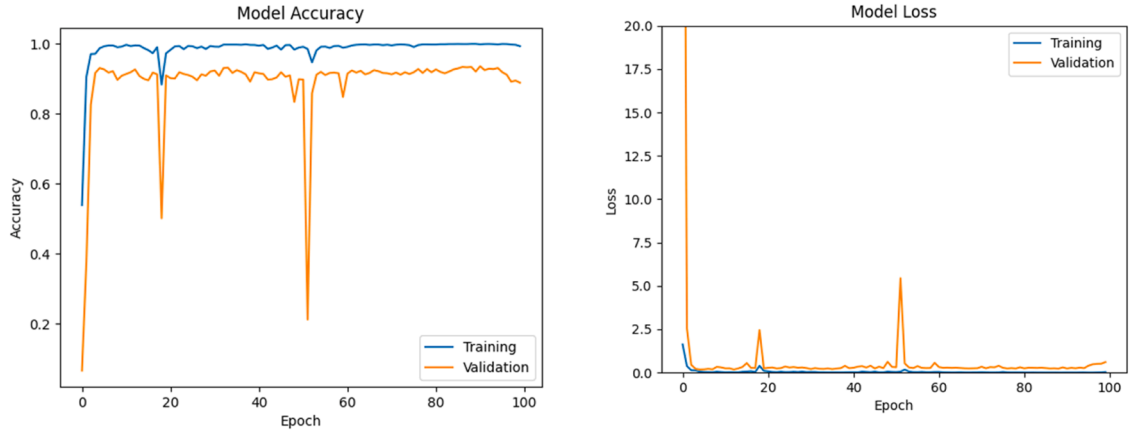


Fig. 19. Leak localization model architecture.



(a) Model training and validation accuracy. (b) Model training and validationo loss.

Fig. 20. Training and validation graph of accuracy and loss.

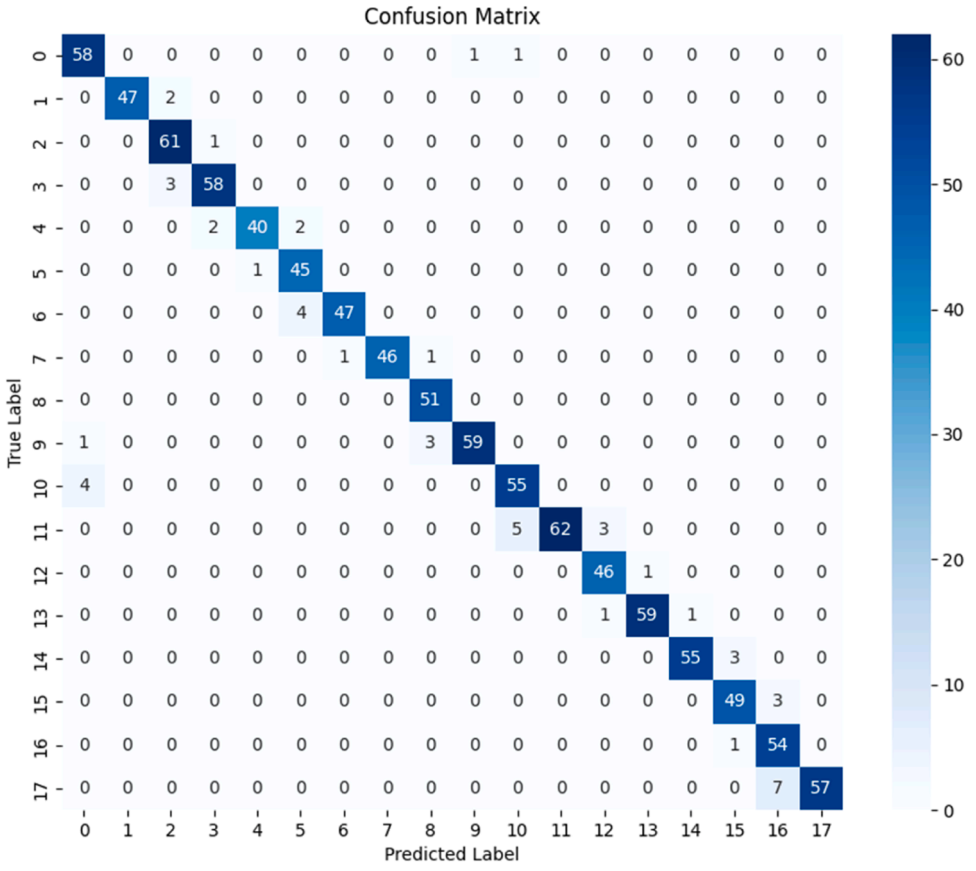
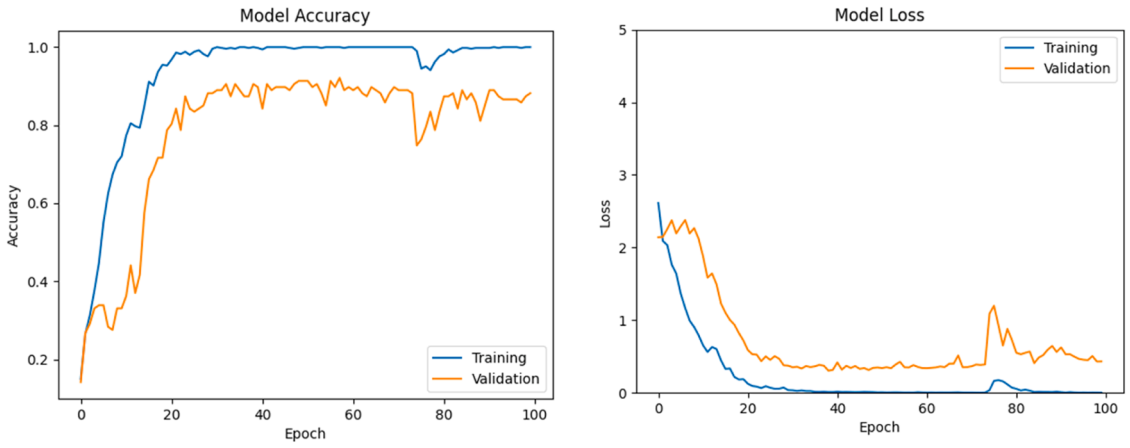


Fig. 21. Confusion matrix.



(a) Model training and validation accuracy. (b) Model training and validation loss.

Fig. 22. Model accuracy and loss curve.

that out of 127 testing images our model could correctly classify 112 images. It meant that our model had a 88.19 % accuracy.

Table 10 shows the performance comparison of the base model with CAMIL dataset and after transfer learning with captured dataset. The base model's performance was before transfer learning.

The base model had 18 classes but as we had limited data available, we changed it to 9 classes. The model had 94.81 % accuracy with CAMIL dataset and 88.19 % accuracy with the test dataset. The specificity of both models is close but apart from that all the other parameters are approximately 7 % lower in the transferred learned model.

7. Limitations

Our investigation yielded valuable insights into the detection of leaks through the utilization of thermal images and geophones.

Environmental factors such as sunlight, tree shade, and ambient temperature changes could affect thermal images. These changes could result in false positives. The system might fail to detect small leaks or leaks in areas with poor thermal contrast, like in cold climates with weak sunlight. The system could be most helpful in tropical and hot climates, where temperature transitions day and night. The transition of day-to-night or night-to-day gives the most contrast in the thermal soil

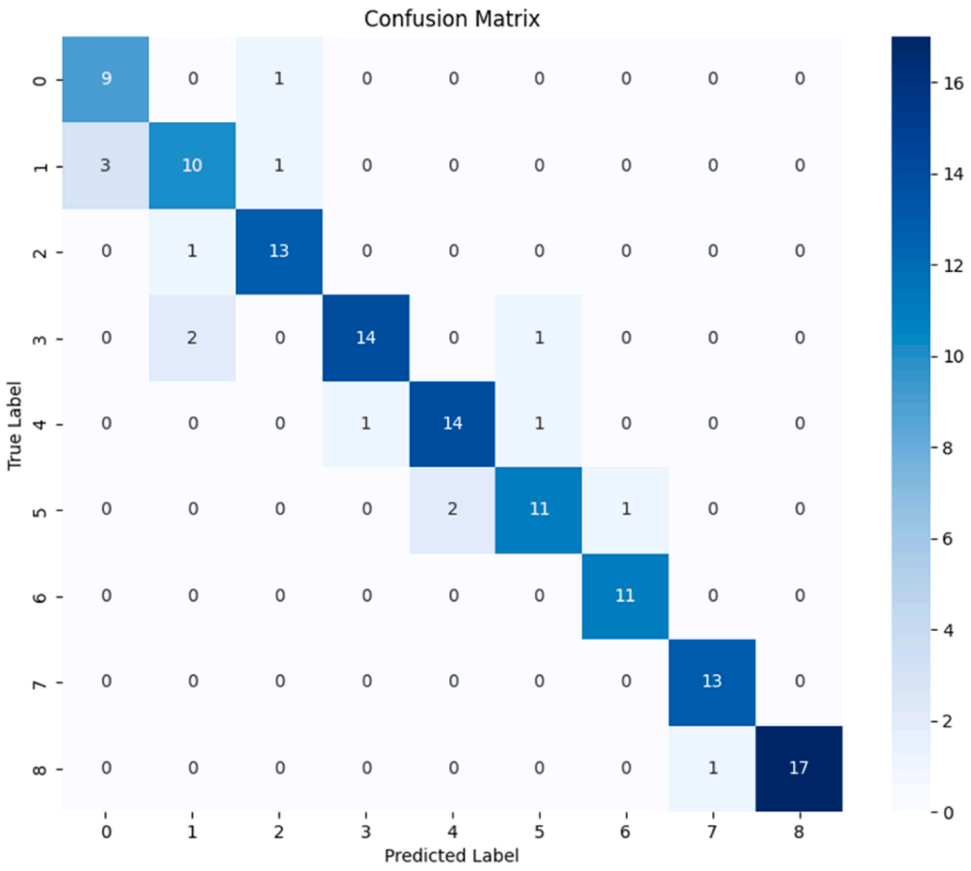


Fig. 23. Confusion matrix.

Table 10
Performance comparison.

Model	Accuracy	Precision	Sensitivity	Specificity	F1-score
Proposed model with CAMIL dataset (Before transfer learning)	0.9481	0.9500	0.9495	0.9970	0.9485
Proposed model after transfer learning	0. 8819	0.8762	0.8857	0.9853	0.8790

images. False reading could be minimized by taking multiple readings of the same place.

On the other hand, Geophones could not pick up small leaks when they did not emit sound. Equipment noise could interfere with the leakage sound. It could not work near heavy traffic or heavy operating equipment. Dry soil is not a good sound conductor and conductivity depends on the moisture content so the probe could not pick up underground leakage noise from long distances (Brutsaert & Luthin, 1964). In our experiment, the maximum range was approximately 1 meter. As soil is not a homogeneous material, two probes on different soil types could impact the device's effectiveness. Soil saturation conditions could also impact the device's performance, as sound travels more quickly in saturated than dry soil.

8. Conclusion and future work

We presented a novel way to detect and localize water leakage. It demonstrated the ML's potential in water leakage detection. It was a two-step process that involved using thermal images and stereo geophones to pinpoint leakage in an area. The process was beneficial for surveying large urban areas.

The first one is thermal image based. We used image processing and ML to create the network. The thermal image-based leak detection network was a hybrid network that combined convolution, soft-attention, and LSTM layers. The network showed an accuracy of 95.20 %. In the previous research, the thermal image-based leaked detection method required human translation and was not automated. Our proposed system was fully automated and could handle a large number of images in a very short time.

For leakage localization with sound, we used two standard features of stereo audio, ITD and IPD. Our dataset was small, so we utilized transfer learning to offset this problem. At first, we used the CAMIL dataset to train and optimize an ML model and later used that model again with the field dataset. The ML model was a 1D-CNN model. The final model performed very well, reaching 88.19 % accuracy. Considering the errors in degrees, it is within the ±20-degree range. Multiple positions could be probed to triangulate the underground leakage location.

The section describes how we intend to use the two proposed ML models in real life. The proposed underground water leakage detection is a two-stage process. In the first stage, we could use thermal images for approximate localization, and in the second stage, we could use geophones to get precious locations.

Residential water supply pipes are laid along the road's verge or footpath. First, a car or a drone could capture thermal images alongside the verge or footpath. It needed to capture at least two sets of thermal images for each location. The first set could be captured early in the morning or late in the afternoon when there is no direct sunlight on the ground. In these times, the saturated soil has a significant temperature difference from their surrounding soils and is visible in the thermal

images. The second set of images could be captured after direct sunlight on the ground for a few hours. This direct sunlight removes small moisture build-up like dew from the ground, but the thermal signature of the leakage location would still be present in the ground. After capturing the images, it should be run through the ML models. If the model detects leakage in both images of a particular location, it could be classified as a leakage location. Thermal-based water leak detection would give us an approximate water leakage location. Geophones could be used to locate the leakage location preciously.

Our proposed solution could be vital in curbing water wastage and promoting resource conservation. Wide-scale adoption of the proposed system in municipalities could reduce leak detection survey time and reduce non-revenue water losses. Reducing water loss offers economic benefits by driving cost savings and efficient resource allocation. Moreover, its environmental impact could be significant, as it could drastically reduce energy consumption for water transport, resulting in lowered carbon emissions and reduced ecological stress. When implemented on a large scale, the proposed solution could be a powerful catalyst for positive change across various aspects of human life.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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